A National Airspace System Reference Architecture as a System of Systems

Tracing Operational Intent from Enterprise Objectives to Aircraft Trajectory

ISD 503 Capstone Project Report – Interim Report #2

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Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section 6), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report. State the two claims here as well as in the Introduction (Section 1; owner's note on the 2026-10-11 print, page 11).

- **Problem.** NAS decisions are made by independent actors, each with partial information. When a conflict or disruption occurs, the decision is made quickly and myopically. The sector controller resolves conflicts knowing type, level, speed and route, but not the aircraft's performance.
- **Gap.** Few sources model the NAS with systems engineering. None found address what detailed aircraft state information is worth to the decision.
- **Claims.**
  - Experiment: a higher-fidelity decision aid, fed accurate performance data and the wider airspace state, changes how conflicts are resolved.
  - Model: an MBSE architecture locates the missing interface, names who owns the decision and the data, states the unmet requirement, and lets the scoring fidelity be swapped with the structure unchanged.
- **Approach.**
  - SysML v2 model of the NAS as a system of systems: context diagram, internal block diagrams, nominal-operations activity, requirements, verification cases, the proposed airline-to-ATC interface.
  - Experiment: one level-crossing conflict, four aircraft, one sector; four cases (A to D) with increasing information; options scored with OpenAP at three fidelities; impacts reported side by side for aircraft, airline, sector and system.
- **Findings: architecture.** [Measures from Section 5: requirements traced, requirements allocated, requirements tagged `improved`.]
- **Findings: experiment.** [Option chosen by each case; B minus A (value of automation) and C minus B (value of the information exchange) in fuel, CO2 and instructions; who pays.]
- **Impact.** [From Section 6: operational and dollar terms where a credible conversion exists.]
- **Recommendations.** [From the Conclusions and Recommendations section.]

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1 Introduction

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

This project builds a reference MBSE architecture for how airspace intent is managed across the National Airspace System (NAS). The architecture spans the organizations that define and enforce rules, the domains that manage airspace resources, the systems that support flight operations, and the onboard systems that turn intent into an executable trajectory and guidance behavior.

DeLaurentis [16] states the problem as follows:

> "Problems of significant complexity are facing decision-makers within government and industry, and the moniker of 'system-of-systems' (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the 'stovepipe' mode – their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems ... A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time."

The statement dates from 2005, but still describes the NAS today. The NAS is a system of systems, and the decisions its stakeholders make have consequences for the safety, efficiency, and sustainability of air transportation.

Airlines invest significant time and money in operational plans that maximize revenue. On the day of operation, unpredictable events – bad weather, aircraft malfunctions, crew absenteeism – can substantially alter that plan and the revenue objectives behind it [11]. To manage this, airlines maintain an Airline Operations Control Center (AOCC), a department of specialists in disruption management across aircraft, crew, and passenger recovery. The AOCC monitors operations and makes real-time decisions – re-routing flights, rescheduling crews, rebooking passengers – intended to limit the effect of a disruption on the airline's schedule and revenue.

The literature does not converge on one standard definition of the "cost" the AOCC is minimizing, and the values it does quantify do not always hold up under scrutiny. The optimization literature commonly models delay cost as a linear function of delay minutes, but Hu et al. [46] note that this is inconsistent with airline practice, where delay cost is a nonlinear function of delay time, and that, faced with this complexity, airlines tend to put the speed of a recovery decision ahead of the mathematical optimality of the solution. The same preference for fast, tractable decisions shows up in how the work is structured. A review of the aircraft recovery problem finds that studies – and, by extension, the AOCC processes they model – rarely minimize delay, cancellations, and aircraft swaps together. They treat aircraft, crew, and passenger recovery as separable sub-problems, solved in sequence and handed off in turn [79]. Each stage can produce a solution that is locally optimal yet globally suboptimal, or even infeasible, once the next stage's constraints are applied. This is a documented instance of the myopic-optimization pattern this capstone investigates.

The pattern is visible even in work that improves on the status quo. The predictive turnaround-buffer allocation model of Kontodimou et al. [58] achieves a 37.1% reduction in propagated delay

relative to a zero-buffer baseline, but it reaches that result by holding aircraft routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as upstream constraints, outside the optimization. The gain is real within that scope. The framing also shows the risk this capstone is concerned with: a locally optimized decision layer can report a strong local benefit while passing tradeoffs to adjacent decision layers that its own cost model does not measure. The same pattern appears on the human side of the AOCC. Turhan, Gzara, and Elhedhli [91] document that when dispatcher workload exceeds capacity, dispatchers "may resort to conservative approaches, which can have financial implications for airlines" – a workload-driven cost that rarely appears in the delay, cancellation, and swap cost models discussed above. Hassan, Santos, and Vink [38] frame this more broadly as a persistent gap between the disruption-management optimization literature and AOCC practice.

This report explores the value tradeoffs aviation stakeholders experience when a disruption occurs and the recovery decision is made with more or less myopia. A dispatcher or AOCC team member making a decision is often focused on its immediate effect – for example, minimizing delay for one flight – without visibility into what it means for the airline's wider operation or for the NAS as a whole. In this view, the aircraft is the top-level system of interest being considered. The cost models in use today are over simplifications that capture the factors easiest to quantify, such as fuel and crew cost. Factors that are harder to quantify – maintenance cost, passenger satisfaction, dispatcher and controller workload, environmental impact – are more often left out. The result can be congestion in particular airspace sectors, missed opportunities for more efficient routing, or lower profit driven by customer dissatisfaction. By modeling the NAS as a system of systems and tracing how operational intent propagates through it, the project aims to identify ways to improve these decisions and reduce the cost of disruptions to the air transportation system.

Two claims are central to the project. The first, explored by experiment, is that replacing a low-fidelity rule of thumb with a higher-fidelity model fed by accurate aircraft performance and informed of the broader state of the airspace can and will lead to changes in how disruptions are mitigated. The second, shown by modeling, is that an MBSE architecture can locate that interface, name who owns the decision and the data, state the unmet requirement, and lets the scoring model's fidelity be swapped (conceptual, nominal mass, actual mass) with the structure unchanged. As such, the model contributes as a traceable representation of how strategic objectives become executable aircraft behavior.

1.1 Overview of the Project

In the NAS, operational decisions are made by independent actors, each of which may or may not account for the effect on the others. Those actors include regulatory agencies, air traffic service providers, airport operators, airline operations centers, maintenance (MRO) suppliers, flight crews, passengers, and the aircraft systems themselves. Substantial architectural work exists within individual domains, but the relationships and information flows between domains are often incompletely represented.

Existing models frequently focus on one actor or one problem, and show one or more of the following limitations:

- emphasis on enterprise-level capabilities without sufficient representation of aircraft behavior and onboard decision-making;
- emphasis on aircraft systems without adequate representation of the operational context in which those systems operate;

- limited traceability between strategic objectives, operational constraints, human decisions, and the resulting aircraft trajectories;
- difficulty evaluating the effect of new operational concepts, automation technologies, and optimization services across organizational boundaries.

As a result, no broadly applicable architecture fully captures how strategic objectives become executable aircraft behavior. Without end-to-end traceability, engineers, operators, and decision makers have limited means to understand the consequences of a change within the NAS, and integrating future decision-support technology is harder than it needs to be.

This project addresses that gap by developing a System-of-Systems architecture that explicitly models the propagation of intent through the NAS, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

1.2 Literature Review

A literature review explored the makeup of the NAS more generally and the various problem spaces that it contains. The identified literature can be categorically organized as follows:

- Long-horizon strategic airline planning and operations
- Day-of-operations and collaborations necessary to begin a flight's execution
- Tactical flight execution and disruption management

Airline planning and operations literature focused on schedule planning, fleet assignment, crew scheduling, and maintenance routing. In general these operations consist of a series of optimization problems that are solved sequentially, long before a flight enters the NAS, and the chosen solutions define the input conditions to the day-of-operations and turnaround, and OCC/dispatch and flight execution problem spaces, which are more directly relevant to the intent-to-trajectory problem space this capstone addresses.

The literature focused on day-of operations explored how airlines and airports coordinate to manage flight schedules, gate assignments, and turnaround times, as well as how weight and balance considerations affect aircraft performance and safety. The OCC/dispatch and flight execution literature focused on disruption management, dispatcher responsibilities, and nominal ATC/IFR flow.

Very few of the sources utilized a systems engineering approach to model the problem space and the relationships between the various actors and systems. [60] developed a model of the NAS using MBSE to represent the various actors and systems, but it did not focus on the propagation of intent through the system and did not publish their models publicly.

The EU Greener Air Traffic Operations (GREAT) project [45] developed a model of the NAS using MBSE to represent the various actors and systems, which was the most comprehensive found. On page 37 the authors explore the modifications made to a flight throughout the cruise phase and how those are communicated and coordinated between the various actors and systems. However, the GREAT project did not address the value of detailed state information and the effect of that information on decision-making and trajectory execution.

The lack of modelling is a gap in the existing research, which this capstone addresses by developing a System-of-Systems architecture that explicitly models the propagation of intent through the NAS, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

The gap represented by the lack of information exchange and the effect of that information on decision-making and trajectory execution is addressed by the experiment, which evaluates the effect of providing additional information to the ATC sector controller on the resulting trajectory and its associated costs. [15] measured what airline-supplied weight does for climb prediction. Much work has explored the potential value of the Extended Projected Profile (EPP), which will provide for downlink of the flight management system's (FMS) predicted profile, but it has not been widely implemented and it does not provide for what-iffing.

Furthermore, the experiment explores the impact of how the controller defines optimality when resolving a conflict between two aircraft. This project bridges these gaps by:

- scoring one decision at four stakeholder levels side by side (aircraft, airline, sector, system) with no weighted sum, so who pays is visible;
- mapping the decision to its authority and its information owner in a system-of-systems model, so the missing exchange is a named interface and an unmet requirement, not a finding in prose;
- swapping the scoring model's fidelity under a fixed architecture and measuring its impact, which is the MBSE claim.

In the moments when the conflicts arise and disruption management must occur, today's controllers have type, level, speed and route, but not the aircraft's performance. [81] describes how weight and speed intent "are considered competitive parameters by airlines and are not available for use". As such, the what-iffing aspect of the additional interfaces must be executed carefully and with security in mind.

[75] describes how this lack of information leads to simplified solutions, where controllers "use more predetermined rules", leading to 23 of 36 level-crossing cases resolved by a descent. This simplification is further supported by the findings of [57], whos investigations determined that controllers "leave the aircraft in steady state alone, and by [35] who quoted one controller as stating there's "No point asking aircraft to climb if they come back and say they can't".

[19] explored the value of providing traffic information to the flight deck where it can be combined with performance data to determine optimal trajectories, and found that it led to a reduction in fuel burn and time for the typical case. This project is a mirror-image study, where the traffic picture is provided to the controller instead of the flight deck, and the effect on the trajectory and its associated costs are measured. The study did not model aircraft weight and limited climbs to FL350 and below for that reason. This is exactly the assumption Section 4.2 adopts for a point of comparison.

1.3 Stakeholders and Study Boundary

Informed by the literature survey and 16 years of experience in the aviation industry, a broad set of candidate stakeholders was identified with a PESTLE analysis (Section efsec:stakeholders). Four are kept: operators (the airline and its operations center), air traffic control, the airport authority, and the flight crew. The ommitted stakeholders supply rules, equipment, demand, and expectations to the decision but do not make it or execute it. As such, the governments and standards organizations, aircraft and engine manufacturers, passengers, and environmental and societal organizations are not explicitly modelled. Section 3 gives the full stakeholder analysis (Section 3.1), the reason for each choice (Section 3.1.3), and the resulting System of Interest (Section 3.3).

The study examines one tactical decision, the resolution of a conflict between aircraft in cruise, and keeps only the stakeholders and objectives that bear on it. Six objectives are kept: safety, efficiency, environmental impact, schedule reliability, cost, and workload. Throughput and capacity

are left for another study, and the phases before departure enter only as the conditions the studied decision inherits.

1.4 Project Deliverables

The project delivers the eight items of Table 1.

Table 1: Project deliverables.

Deliverable	What it shows	Where in this report
Context diagram	The systems kept in the System of Interest, the actors at its boundary, and the exchanges among them	Section 3.3
Block definition diagrams	The NAS broken down into domains and constituent systems	Appendix A.4
Internal block diagrams	The parts inside the aircraft and inside airspace management, and the links among them	Section 3.5
Activity diagrams	One nominal flight from gate to gate, and the place of the conflict within it	Section 3.4
Requirements	Stakeholder requirements, system requirements and constituent-system requirements, each traced to the tier above	Appendix A.4
Experiment	One conflict resolved four ways and scored for the aircraft, the airline, the sector and the system	Sections 4 and 5.2
Traceability chain	One thread from the airline's cost objective through the requirements and the airline-to-ATC interface to the experiment's outcome table	Section 5.2; Appendix A.4
Gap analysis	The requirements that today's NAS does not meet, and what the experiment measures for each	Section 5

1.5 Project Timing

The plan established at the project's inception is presented in Table 2 for reference, with the status of each item: not started, draft, complete, or will not implement.

Table 2: Project milestones and verification methods of the original plan, with the status of each.

Project Milestone	**Deliverable**	**Lead/Team Member**	**Expected Completion Date**	**Target Metric/Verification Method**	**Status**
Literature Review	Annotated bibliography	Owen	9/7/2026	10–20 papers categorized	Draft
PESTLE Analysis & Systems-of-Interest Definition	Context diagram, text-based documentation	Owen	9/10/2026	Review	Complete
Systems-of-Interest Definition	Text-based documentation	Owen	9/15/2026	Clear boundaries and exclusions	Complete
Operational Scenarios	ConOps document	Owen	9/22/2026	1–3 representative scenarios	Draft
Draft Architecture	BDDs and package structure	Owen	10/1/2026	All domains represented	Complete
Interface Definition	Interface Control Document	Owen	10/12/2026	All interfaces identified	Draft
Internal Block Diagrams	IBDs	Owen	10/17/2026	Critical interfaces connected	Draft
Requirements Definition	Requirements model	Owen	10/20/2026	Trace to stakeholders	Draft
Measures of Effectiveness	MOE/MOP hierarchy	Owen	10/25/2026	Quantifiable metrics defined	Complete
Requirement Traceability	Requirements matrix	Owen	10/29/2026	Complete allocation	Draft
Activity Diagrams	Operational activities	Owen	11/3/2026	Scenario walkthrough	Draft
Sequence Diagrams	Information exchanges	Owen	11/8/2026	Message completeness	Not started
State Machines	State behavior of key entities	Owen	11/13/2026	Executable state logic	Will not implement
Authority & Responsibility Models	RACII (Responsible, Accountable, Consulted, Informed)	Owen	11/17/2026	All decisions assigned	Will not implement
Optimization Prototype	Working code	Owen	11/21/2026	Unit test cases	Draft
Integration into MBSE Architecture	Traceability links	Owen	11/24/2026	End-to-end scenario	
Final Baseline & Final Paper	Versioned model & thesis/paper	Owen	12/16/2026	Draft	Not started
Paper Submittal	Thesis/paper	Owen	12/20/2026	Official submittal	Not started

2 Assumptions and Methodology

The project uses a model-based systems engineering (MBSE) approach to define the architecture and the decision-making process, and a simulation-based experiment to test the proposed updates to them. First, a literature review was conducted to:

1. identify the current state of air traffic management,
2. identify the relevant stakeholders and their objectives,
3. identify the requirements necessary to meet those objectives, and
4. identify the gaps in the current system that the proposed interface and decision-making process can address.

The contextual information captured in the literature review enabled the execution of the systems engineering approach.

2.1 Systems Engineering Approach

Standard systems engineering methods of analysis and decomposition were applied to the collected evidence, which produced a context diagram, internal block diagrams, and activity and sequence views. These identified the relevant systems and the interactions among them that are necessary to meet the stakeholders' objectives. From them the stakeholder requirements were written and then decomposed into system-level requirements (SLRs) for the NAS. The SLRs were allocated to the relevant constituent systems, and high-level requirements (HLRs) decomposed them one level further.

This enabled the project to (1) build a model of the architecture and the decision-making process, and (2) compare current air traffic management practice, which relies on general rules of thumb and knowledge of what has worked before [75, 57], with a model-based decision-support approach that replaces the low-fidelity decision rule with one or more higher-fidelity models of aircraft performance, at varying levels of accuracy and information.

The emphasis is on structure, behavior, interfaces, and traceability, not on optimization mathematics. The ISO/IEC/IEEE 15288 technical processes were followed in a tailored form, which the standard provides for.

Table 3 summarizes the systems engineering processes and the artifacts that realize them.

2.2 Working Assumptions

The working assumptions fall into two groups: assumptions about the problem and its context, and assumptions about the methodology.

The problem statement is that the NAS is hard to change, and that a new airline-to-ATC interface and a new way of deciding which aircraft to move in a conflict could lower the cost of air traffic management. This framing assumes that:

- the abstractions used in the model are sufficient to capture the relevant behavior of the system;
- the stakeholders' objectives are comprehensively captured and accurately represented in the model;
- the scenario is sufficient to explore the potential benefit of interconnected MBSE.

Table 3: Systems engineering processes and the artifacts that realize them.

Process (15288 / INCOSE)	Artifact in this project	Where
Stakeholder needs and requirements	30 stakeholder requirements (11 in the study), each traced to an objective row	`requirements_stakeholders.sysml`; Appendix A.4
System requirements	42 system requirements (23 in the study), allocated to constituent systems; 31 constituent-system requirements one tier down	`requirements_nas_system.sysml`; `requirements_constituent_systems.sysml`
Architecture definition	Context diagram, internal block diagrams, typed interfaces on the thread, activity and sequence views	`cameo_models/`; Section 3.5
System analysis	Experiment at three fidelities, outcome table	`simulation/`; Sections 4 and 5.2
Verification	Verification cases for the improved requirements, run by the experiment	`nas_verification.sysml`
Validation	Stakeholder-level reading of the outcome table (who pays) with MoEs or MoPs	Section 5

Every abstraction was made with the express goal of capturing the relevant behavior of the system, as defined by the stakeholders' objectives. The System of Interest, the NAS as a system of systems, was ratified around the trajectory-intent lifecycle, from enterprise objectives through aircraft motion. An entity is modeled when it has distinct decision authority or execution behavior that is dynamically coupled to in-scope decisions and materially affects cost, safety, workload, schedule reliability, or passenger value. Boundary actors supply objectives, demand, feedback, or rules without being decomposed as decision mechanisms. Exogenous influences enter as constraints or scenario parameters. This criterion preserves the authority and responsibility relationships a system-of-systems architecture needs without expanding the project into a model of every NAS participant.

The methodology is a model-based systems engineering approach that uses simulation to test the proposed interface and decision-making process. It assumes that:

- models of every required aircraft type are available;
- the security concerns of sharing airline data can be mitigated;
- the proposed interface can be implemented in the NAS over digital communication channels between the airline and ATC, between the aircraft and ATC, or both;
- the airline reports its aircraft performance accurately;
- the conflict can be treated as a snapshot, with the traffic unchanged while the decision is made;
- the specific scenario presented in the experiment is representative of a larger set of scenarios that generalize to the NAS as a whole.
- wake turbulence is not modeled, so the aircraft are assumed to be separated by more than the minimum wake-separation distance.

The explored scenario stands for a narrow set of conflicts: those between two aircraft crossing at the same level. Rantanen and Wickens [75] found that this represented only 36 of 256 conflicts

identified in the operational data they studied. The scenario is represented through one hypothetical en-route sector and four aircraft, with controllers responsible for maintaining separation and for resolving conflicts. The aircraft are assumed to be at the same altitude, with the same speed, and on a collision course. The scenario is designed to be simple enough to allow for a clear analysis of the proposed interface and decision-making process, while still being representative of the types of conflicts that occur in the NAS.

The controllers are assumed to have today's level of information about the aircraft and their intentions, and are equipped with varying levels of information and automation. In the baseline scenario, the controllers have only the information that is currently available to them in the NAS. In the proposed scenario, the controllers have access to additional information about the aircraft's performance, which allows them to make more informed decisions about which aircraft to move in a conflict. Inherent in this assumption is that the reported state information is accurate ([59] discusses how airlines can game the system with inaccurate information) and that the underlying performance models are sufficiently accurate to support the decision-making process (its simplifications are listed separately in Section 4.4). The controllers are also assumed to have the ability to communicate with the aircraft and the airline, and to be able to issue instructions to the aircraft in a timely manner.

3 System Model

This section presents the identification and formulation of the model the experiment is built on, beginning with the stakeholders, their objectives and the conflict among them. It then identifies the constituent systems through which they act, and draws the boundary of the System of Interest. Finally, it describes how a flight operates within that boundary, the architecture that carries it, and the airline-to-ATC interface whose use informs the decisions that determine the value achieved by the experiment of Section 4.

3.1 Stakeholders

3.1.1 Identification

Informed by the literature survey and the author's experience in the aviation industry, a set of candidate stakeholders was identified and refined using a PESTLE analysis and stakeholder discovery process. The result is a diverse group of stakeholders that represent the various interests and perspectives affected by NAS operational decisions, and is shown in Figures 1

The stakeholders are grouped by the six domains of a PESTLE analysis (political, economic, social, technological, legal, environmental) and scored for importance and impact on trajectory decisions in 2. An initial assessment of each candidate stakeholder's objectives, candidate MOPs and MOEs, and whether the stakeholder has a direct or indirect effect on trajectory decisions is documented in the `capstone_stakeholder_assessment.xlsx` file. The assessment also considers whether the stakeholder should stand alone or be abstracted through consolidation with another, and the reason for that decision. An excerpt of the work is in Appendix A.2, Table 10.

The power analysis informed the decision to keep four stakeholders in scope: the airline, air traffic control, the airport authority, and the flight crew. The other stakeholders are treated as boundary actors that supply rules, equipment, demand, and expectations to the decision but do not make it or execute it. The four kept stakeholders appear on the context diagram (Figure 4) and are decomposed in the model.

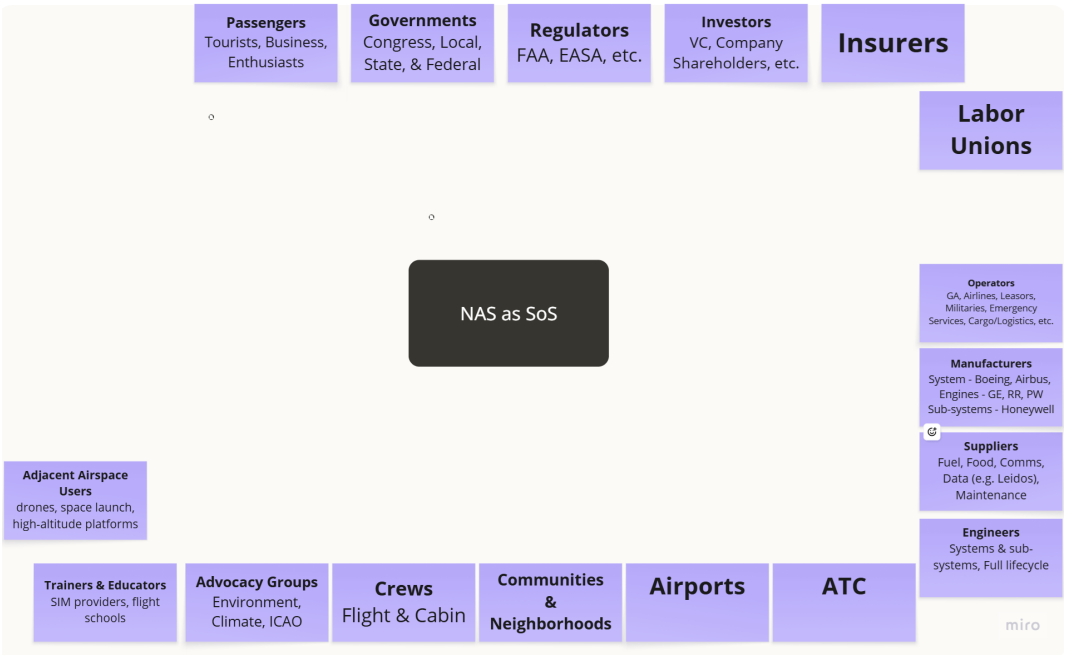


Figure 1: Candidate stakeholders of the NAS as a system of systems.

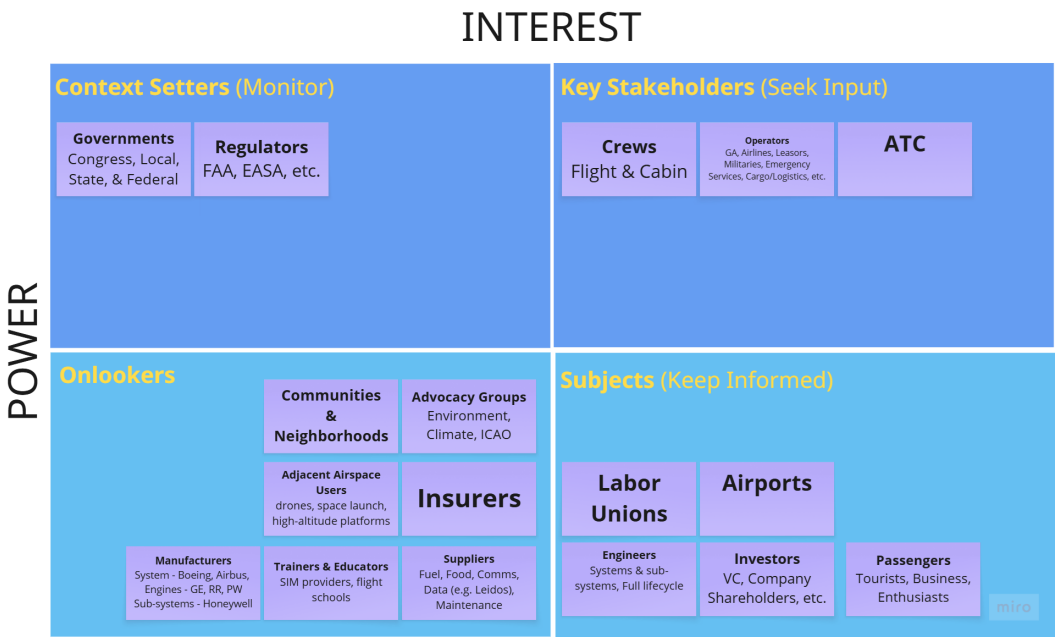


Figure 2: Candidate stakeholders placed by power and interest.

3.1.2 Candidate Objectives

An exploration of each stakeholder's interests led to the following list of objectives relevant to the project:

Each item is an objective because at least one stakeholder seeks to affect it, and it can be measured, as a measure of performance for one system or a measure of effectiveness for the whole.

Safety / Security Keeping aircraft separated from each other, from terrain and from hazardous weather, and protecting the operation from deliberate harm. Held by every stakeholder; for air traffic control and the flight crew it is a constraint that is never traded.

Efficiency Flying the flight with the least fuel and time for the distance covered. Held by the airline, whose fuel cost depends directly on the trajectory flown and whose time cost is determined by the passengers and the number and seniority of the crew conducting the flight.

Throughput / Capacity The number of aircraft a sector, runway or airport can handle in a period. Held by air traffic control and the airport authority.

Reliability / Resilience The ability of the operation to run as planned and to recover when it is disrupted. Held by the airline, air traffic control, and passengers.

Environmental Impact The carbon dioxide, other emissions and contrails a flight produces, which follow from the fuel it burns and the path it flies. Held by environmental and societal organizations and, through regulation, by governments.

Noise Impact Sound over the communities near airports, set by the departure and approach paths. Held by communities and the airport authority.

Passenger Satisfaction The passenger's experience of travel time, comfort and connections. Held by passengers, and by the airline through the revenue that follows.

Schedule Reliability Arriving on time and keeping passengers' connections. Held by the airline.

Cost / Profitability The operator's cost of a flight (fuel, crew, the consequences of delay) and the profit it supports. Held by the airline and its investors.

Maintenance / Lifecycle Cost The upkeep of aircraft and engines over their life. Held by the operator and the manufacturers.

Regulatory Compliance Operating within the rules, procedures and clearances in force. Held by the flight crew and the airline, and enforced by the regulators.

Workload / Human Factors The mental and procedural demand on controllers, dispatchers and crews. Held by air traffic control and the flight crew, for whom high workload is a cost and a source of error.

These objectives were then assessed for relevance to the decision under study.

3.1.3 Selection

The stakeholders and objectives above are the result of exploration. The study itself is narrower. It examines one tactical decision, the resolution of a conflict between aircraft in cruise, and keeps only the stakeholders and objectives that bear on that decision.

Four stakeholders remain, together with the aircraft, and each appears on the context diagram (Figure 4): operators, as flight operations and its airline operations center; air traffic control, within airspace management; the airport authority, as airport operations; and the flight crew. The aircraft appears as aircraft systems. Governments and standards organizations, aircraft and engine manufacturers, passengers, and environmental and societal organizations are outside the System of Interest. They supply rules, equipment, demand, and expectations to the decision but do not make it or execute it, so they are treated as boundary actors and not decomposed.

Six stakeholder objectives remain: safety, efficiency, environmental impact, schedule reliability, cost, and workload. Schedule reliability is carried in the experiment as a step in cost when a flight's delay passes a threshold (the parameter K of Section 4.3). The others were set aside for the following reasons:

- Noise impact: the measure is "not relevant at cruise."
- Passenger satisfaction: "not measurable objectively."
- Profitability falls out of the other objectives; it is tracked at the operator's organizational level and is more complex than any one decision.
- Maintenance and lifecycle cost are largely determined by the number of flight cycles, the chosen takeoff performance, the flight distance and the climb controls, so one air traffic control decision has a negligible effect on them.
- Security, resilience and regulatory compliance are not changed by the choice between resolution options in the scenario.
- Throughput and capacity were left for another study. They are also close to ongoing work by the author, and out of an abundance of caution that part of the problem space was avoided.
-

3.1.4 Objectives and Conflicts

Each stakeholder kept in scope pursues a different objective when an aircraft's trajectory is changed. The airline wants the lowest cost for its flights, which combines fuel, time, and the consequences of a late arrival for connecting passengers. Air traffic control wants separation assured with a workload the sector team can sustain (preferrably minimal), and a traffic picture that stays predictable. The flight crew wants instructions the aircraft can fly, given with enough time to set them up.

Some of these objectives align. Burning less fuel generally lowers the airline's cost and lowers carbon dioxide emissions together. Air traffic control's wish for predictability and the airline's wish to keep its schedule both favor aircraft that fly what was planned.

Others conflict, and the conflict this project studies is the first of them: the trajectory that is cheapest for one aircraft is not always the one that costs the sector the least to grant. A request or a resolution that saves fuel for one flight can add instructions, monitoring, and follow-up for the controller. That cost is externalized. It does not appear in the airline's own cost function, and the party that bears it has no means to price it back.

The objectives also operate on different timescales. An airline sets its schedule policy over months, realizes fuel cost per flight, and realizes delay cost in minutes. Air traffic control manages capacity over years through sector design and over seconds through tactical control. A decision made in minutes by one party is judged against objectives the others hold on other clocks.

3.2 Constituent Systems

Stakeholders do not change a trajectory themselves. They act through systems: a controller through the sector's displays and radios, an airline through its operations center and dispatch tools, a crew through the flight deck and the aircraft control mechanisms. And the cost of whatever they decide is actuallized in one place, the aircraft, as fuel burned and time flown. The constituent systems through which the stakeholders act were therefore identified next.

Figure 3 places the NAS at the center and branches outward to the systems that could act on a flight: the aircraft, air traffic control, operators, airports, maintenance suppliers, communication, the regulatory bodies, and passengers. The board is deliberately broad. It follows the system-of-systems view quoted in Section 1, of many heterogeneous systems at several levels, and attempts to list every system that could bear on a flight before any is set aside. Air traffic control divides into the air navigation service providers, whose terminal and en-route facilities control individual flights, and the national command center, whose traffic management units and flow programs shape demand across many. Operators divide into general aviation, airlines and militaries; within the airline, the operations center and its dispatchers sit beside the business, pilots, ticketing and IT functions.

The board decomposes its branches in two different ways, and the difference reflects the problem the project studies. The organizations that decide, air traffic control and the airline, are broken down to the people who make the decision: the individual controller and the dispatcher. The aircraft is broken down to the physical systems that carry it out: the flight management system's guidance and navigation functions, the propulsion components and the fuel system. The decision about a trajectory is made in one part of the system of systems, and its cost is realized in another.

The aircraft is treated alongside the stakeholders as a constituent system. Figure 3 shows the systems first identified within the NAS, with the aircraft decomposed furthest, as its systems and their performance dictate the cost of each flight.

3.3 System of Interest

The System of Interest is drawn from the systems of Figure 3: the systems operated by the four stakeholders kept in Section 3.1.3, and the aircraft. The systems of the boundary actors are not decomposed; they appear only through the context they supply to the decision. The resulting context diagram is Figure 4. It shows the four stakeholders, the aircraft, and the exchanges among them that are relevant to the decision under study.

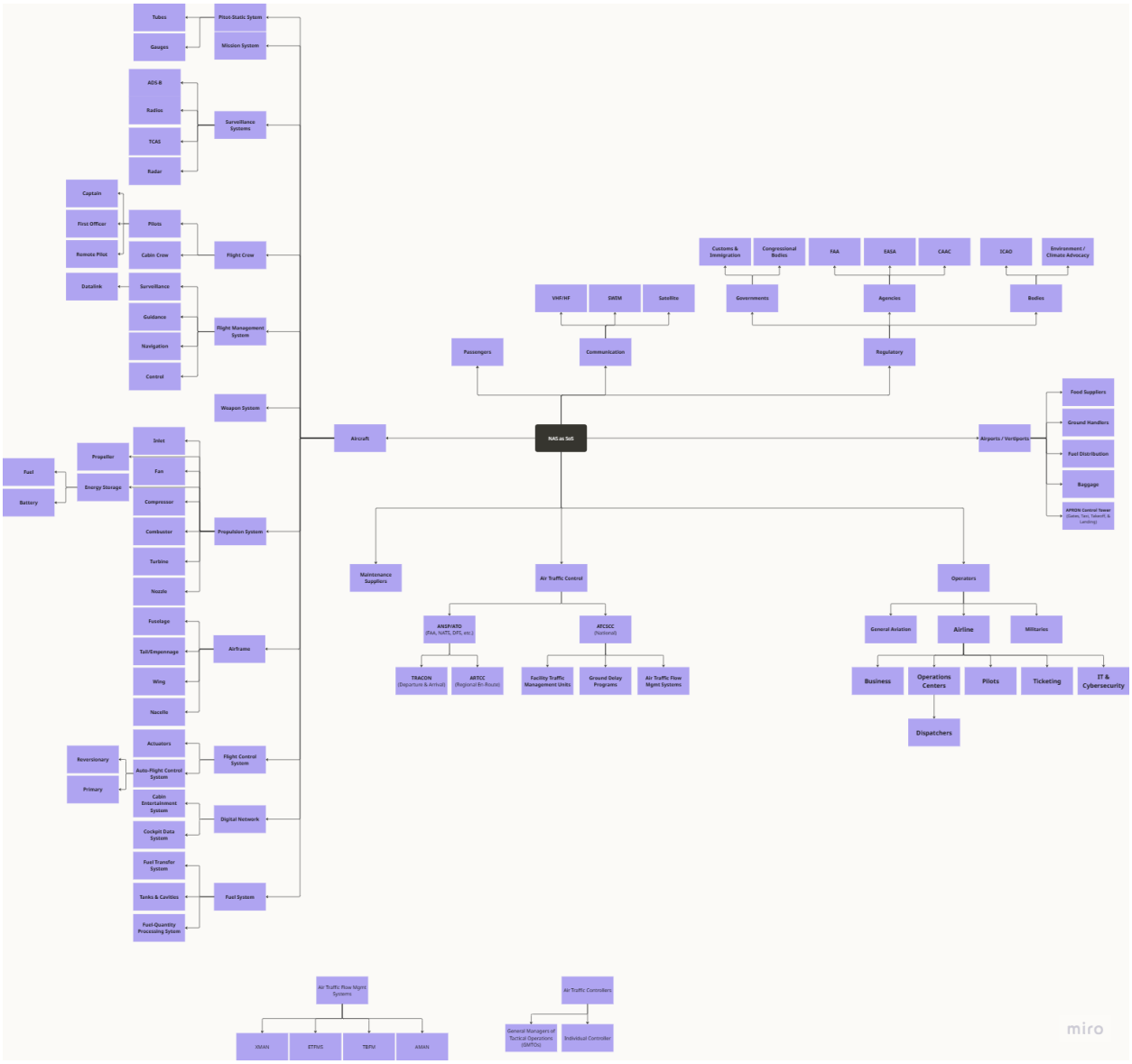


Figure 3: The NAS as a system of systems: candidate constituent systems.

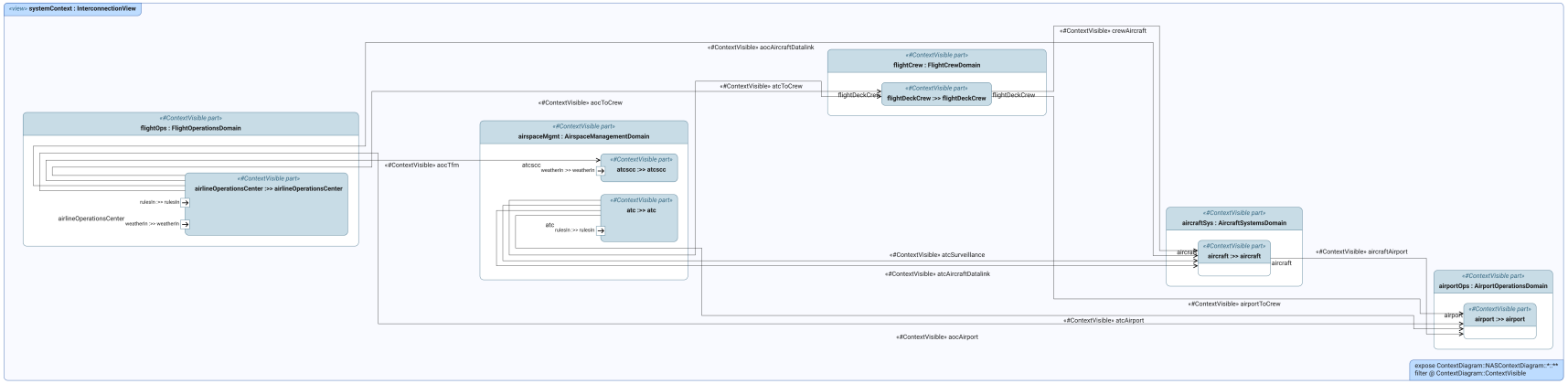


Figure 4: NAS context diagram. View `systemContext`; source `nas_context_diagram.sysml`; rendered 2026-11-05.

Plan ref: to-do-list.md Track C (D-034) — Owner, 2026-10-22: SN-AIR-05 is the throughline the claims are made around, and its decomposition with the improvement HLRs is the output of the report. Write the text that introduces Table 4 (SN-AIR-05, the objectives it serves, SLR-INF-02 and SLR-CLR-13 and their HLRs) and Table 5 (the HLRs today's NAS does not meet), tie them to the context diagram above, and point to the full decomposition in Appendix A.3. Both tables are generated from the model by `tools/export_report_tables.py`. Owner's 2026-11-05 print: the "done today, in part" fact left Table 4 and belongs in this text: before departure, in the FAA's Collaborative Trajectory Options Program, the airline gives each route option a relative cost, which the FAA's allocation uses [59, p. 4]; not in tactical conflict resolution in cruise. Also from that print: SN-AIR-05 needs an interface into the decision mechanism (SLR-INF-02) and the use case itself (SLR-CLR-13).

Table 4: The study's thread: SN-AIR-05, the system requirements that answer it and the sub-system requirements (HLRs) under them, from the SysML v2 model.

Stakeholder requirement	System requirement	Sub-system requirement
SN-AIR-05 The NAS shall take the airline's priorities for a flight (fuel against time, and protected connections) into account when a change to that flight's trajectory is required.	**SLR-CLR-13** The NAS shall choose, among the maneuvers that restore separation, the one with the lowest impact on the aircraft involved.	**HLR-ATC-05** Air traffic control shall choose, among the viable resolution options, the one with the lowest impact on the systems involved. **HLR-ATC-10** ATC decision support shall determine, for each predicted conflict, every resolution option that restores separation and that each aircraft it moves can fly. **HLR-ATC-11** ATC decision support shall determine the impact of each viable resolution option on the systems involved.
	SLR-INF-02 The NAS shall give the party that decides a change to an aircraft's trajectory the information needed to preserve safety and pursue optimality.	**HLR-OPS-05** Flight operations shall make each flight's current aircraft state and performance, and the airline's priorities for it, available to the party that decides a change to that flight's trajectory. **HLR-ATC-08** ATC decision support shall use the best information available to it about each flight, including what the flight's airline shares. **HLR-ACF-05** The aircraft shall make its current state available for sharing with the party that decides changes to trajectories.

3.4 Operational Concept

The operational concept was first written in full detail as a gate-to-gate thread for one nominal commercial flight, from the pilot's point of view. It runs through eleven phases: the airline's schedule, fleet and crew assignment; the pilot's preparation before duty; reporting and the dispatch release; aircraft preflight and initialization of the flight management system; boarding, loading and fueling; clearance, pushback and taxi; takeoff and initial climb; climb; cruise; arrival planning; and descent through to the gate. Each phase records its initial state, the interfaces in use, and the questions it raised. The full thread is in Appendix A.1.

The studied model carries an abstraction of that thread. Figure 5 shows six actions in sequence: plan the flight, prepare for departure, depart, fly en route, arrive, and close out the flight. The eleven phases were reduced to six because the study concerns one decision in cruise. The phases before departure and after arrival fix the conditions that decision inherits (the aircraft's performance, its

Table 5: The improvement: the 5 sub-system requirements (HLRs) that today's NAS does not meet, each traced to its system and stakeholder requirements and the objectives those serve, with how each is verified (the verification definition in `nas_verification.sysml`). The experiment's cases B to D are the means of meeting them.

HLR	Traces to	Objectives served	Verification
HLR-OPS-05 Flight operations shall make each flight's current aircraft state and performance, and the airline's priorities for it, available to the party that decides a change to that flight's trajectory.	SLR-INF-02; SN-AIR-05, SN-CRW-03	Airline: fuel cost, schedule integrity, passenger connections; Flight Crew: operational flexibility, workload, safety; Environmental / Societal: CO_2, NOx, contrail/climate effects, noise, local air quality	By inspection of the case C run: met if, for every aircraft in the conflict, the airline's share (the aircraft's state and performance, and the airline's priorities) reaches the deciding party before the options are scored. `VerifyPerformanceShareProvided`
HLR-ATC-05 Air traffic control shall choose, among the viable resolution options, the one with the lowest impact on the systems involved.	SLR-CLR-13; SN-AIR-05, SN-ENV-01	Airline: fuel cost, schedule integrity, passenger connections; Flight Crew: operational flexibility; Environmental / Societal: CO_2, NOx, contrail/climate effects, noise, local air quality	By analysis, one case per means (A to D): met if the option chosen has the lowest impact (fuel burned and airline cost, per aircraft and summed) among the viable options under the truth model; the four means are compared side by side. `VerifyLowestImpactByMeans`
HLR-ATC-08 ATC decision support shall use the best information available to it about each flight, including what the flight's airline shares.	SLR-INF-02; SN-AIR-05, SN-CRW-03	Airline: fuel cost, schedule integrity, passenger connections; Flight Crew: operational flexibility, workload, safety; Environmental / Societal: CO_2, NOx, contrail/climate effects, noise, local air quality	By analysis, case C against case B: met if the decision support's estimates use the shared values where case B uses the type's nominal ones, so that the options or the choice change where the two differ. `VerifyPerformanceShareUsed`
HLR-ATC-10 ATC decision support shall determine, for each predicted conflict, every resolution option that restores separation and that each aircraft it moves can fly.	SLR-CLR-13; SN-AIR-05, SN-ENV-01	Airline: fuel cost, schedule integrity, passenger connections; Flight Crew: operational flexibility; Environmental / Societal: CO_2, NOx, contrail/climate effects, noise, local air quality	By analysis, case C: met if the options the decision support finds viable are exactly those the truth model finds restore separation and can be flown by every aircraft moved. `VerifyViableOptionsDetermined`
HLR-ATC-11 ATC decision support shall determine the impact of each viable resolution option on the systems involved.	SLR-CLR-13; SN-AIR-05, SN-ENV-01	Airline: fuel cost, schedule integrity, passenger connections; Flight Crew: operational flexibility; Environmental / Societal: CO_2, NOx, contrail/climate effects, noise, local air quality	By analysis, case C: met if the decision support's estimate of each viable option's impact (fuel burned and airline cost, per aircraft and summed) matches the truth model's within a stated tolerance (to be set). `VerifyOptionImpactDetermined`

schedule position) and are kept at the level needed to show where those conditions come from. Detail is added only where the studied decision is made.

The example conflict sits inside the fly en route action. Two aircraft in cruise are predicted to cross at the same level at the same time. The sector controller must move one of them, and each choice costs fuel, time and instructions that fall on different parties. The nominal thread treats a trajectory change in cruise as one step; the experiment of Section 4 opens that step and asks what the decider knows when making it.

Plan ref: to-do-list.md S5 — Done in the model 2026-10-18: every phase is decomposed, `FlyEnroute` is in Figure 6 with `atcResolveConflict` in the ATC monitoring loop, and the other phases are in Appendix A.1. `Prework` was removed 2026-11-05 (D-040), so the figure again has the six actions the paragraph above names. Still to do: reference Figure 6 in the text and check the conflict paragraph against it.

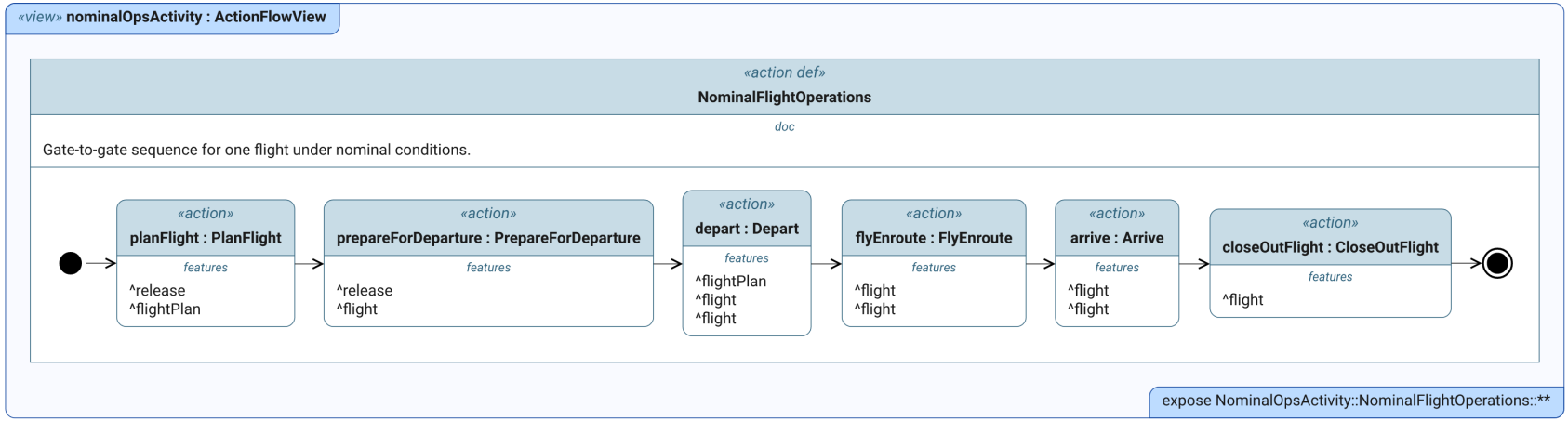


Figure 5: Nominal flight operations activity diagram, phase level (each phase is decomposed in its own figure). View `nominalOpsActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-11-05.

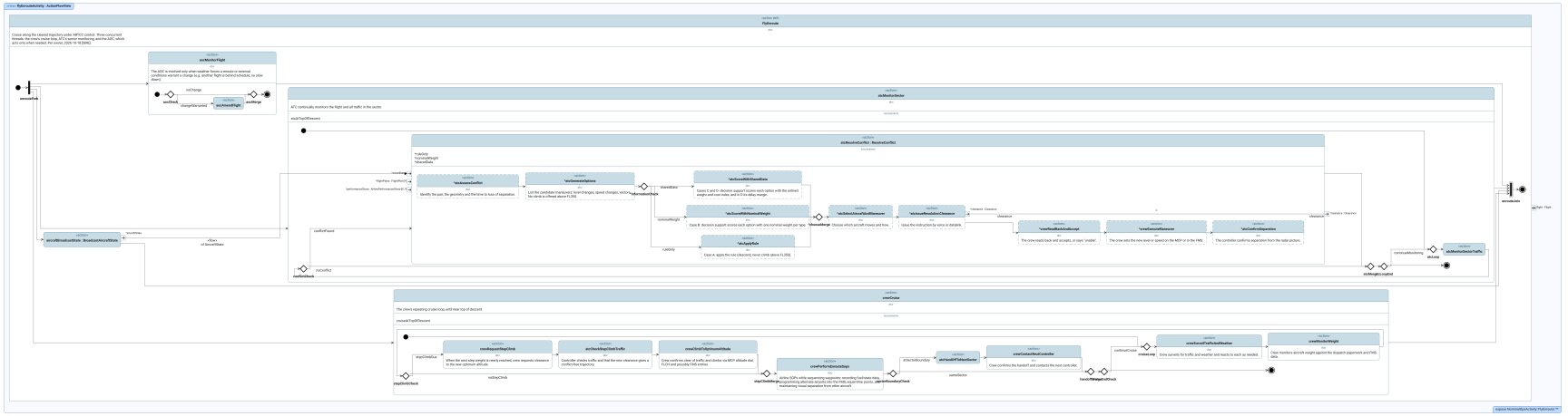


Figure 6: Nominal operations, Fly Enroute: crew cruise loop, ATC sector monitoring and AOC intervention in parallel. View `flyEnrouteActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-11-05.

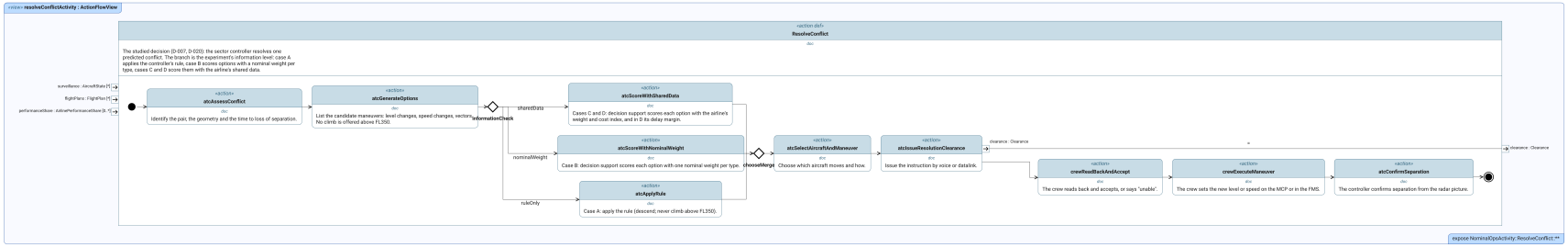


Figure 7: ATC resolves a conflict: the studied decision, branching on what the controller knows (cases A to D). View `resolveConflictActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-19.

3.5 Architecture: Detailed and Abstracted

Plan ref: to-do-list.md S10 — Present the SysML architecture: package structure, the context diagram (Figure 4), the internal block diagrams below, the interface and item-flow model, and the requirements model with traceability. For the aircraft, the detailed diagram comes first and the abstracted one second (your note: "Place detailed and abstracted"); say what the abstraction drops and why. The airspace management diagram is still a placeholder.

The aircraft is shown at two levels of detail. Figure 8 is the detailed internal block diagram, including propulsion, airframe, landing gear, utilities, and the fuel and power links. Figure 9 is the abstraction used in the rest of the report: the crew, the flight deck controls, the avionics, the flight controls, the fuel system, and the information and command links among them, which are the parts the studied decision passes through. The internal block diagrams of the other systems are in Appendix A.4.

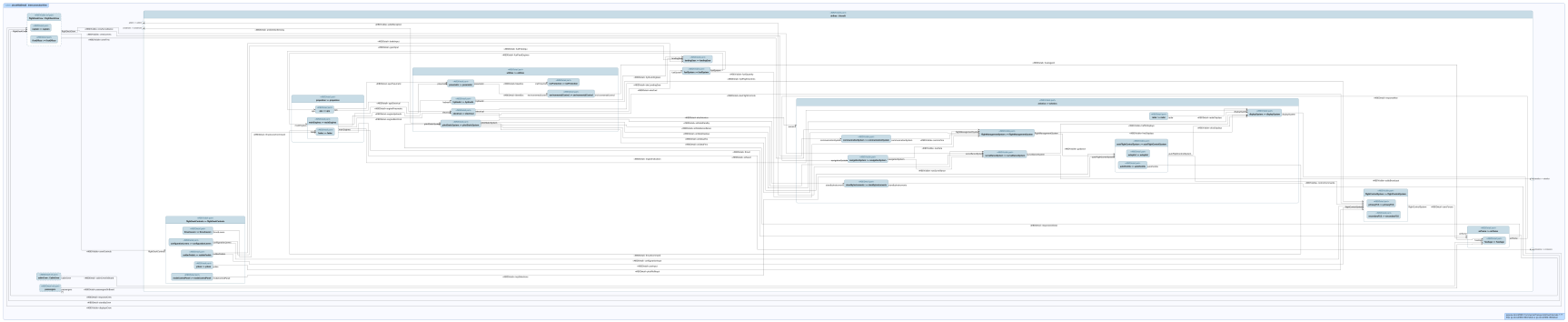


Figure 8: Aircraft internal block diagram, detailed level: adds propulsion, airframe, landing gear, utilities, and the fuel and power links. View `aircraftIbdDetail`; source `nas_ibd_aircraft.sysml`; rendered 2026-11-05.

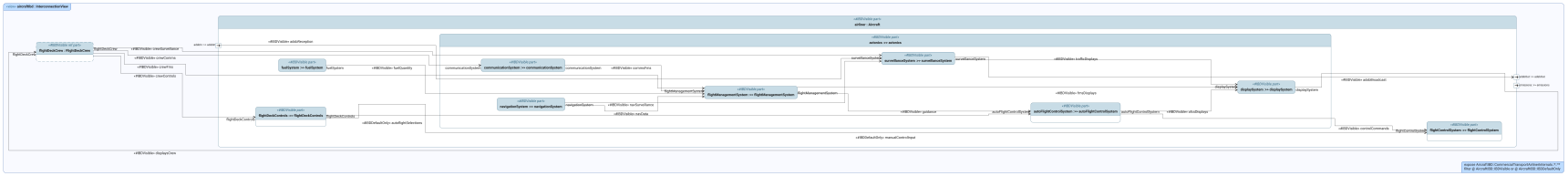


Figure 9: Aircraft internal block diagram, default level of detail. View `aircraftIbd`; source `nas_ibd_aircraft.sysml`; rendered 2026-11-05.

3.6 Proposed Airline-to-ATC Interface

Plan ref: to-do-list.md Track C (D-019) — Name the digital-thread artifact. Describe `AirlinePerformanceShare`: what it carries (the aircraft's state and performance, and the airline's priorities; D-040), who sends it (airline operations center or FMS), who receives it (ATC decision support inside air traffic control, per D-012 owned by the system that decides; Appendix A.4), and which requirements it satisfies (SLR-INF-02, HLR-OPS-05, HLR-ATC-08) and serves (SLR-CLR-13). State that security is an assumption. **Owner's note, 2026-10-11 print: "important to do"** — position it against ADS-C EPP and Coppenbarger's Table 2 (the same items, a different receiver); a first draft of that positioning follows.

The items the interface carries are not new. Coppenbarger [15] listed take-off weight, thrust and drag factors, and the speed profile as candidates for exchange between an airline and air traffic control, and measured what airline-supplied weight does for the prediction of a climb. In Europe, the Extended Projected Profile is sent over ADS-C to downlink the trajectory that the flight management system predicts [84]. What has kept the items out of use is ownership, not feasibility: weight and speed intent "are considered competitive parameters by airlines and are not available for use" [81]. A second obstacle is the availability of first-principles data for each aircraft type: thrust, drag and fuel flow. This is the same obstacle, owned by a different stakeholder. The airline guards its weight and speed intent; the performance data for each aircraft type sits with the owners of the performance models and the manufacturers whose data they draw on. The reference model, BADA, cannot be redistributed, which is why open models such as OpenAP are built from open data instead [90].

The proposal here keeps the concept of information exchange and changes the purpose. In the earlier work the data seeks to make the knowledge of one aircraft's flight path more accurate; here it feeds the decision support of the party that chooses which aircraft to move, thereby providing for asking "what if" within a mathematical framework. The interface is typed in the model, tagged as part of the improved architecture, and tied to the requirement it satisfies, so the report can point at a model element where the earlier work has a conceptual finding. Confidentiality is stated as an assumption: the data is disclosed only to the deciding party, and no security mechanism is modeled.

The experiment that follows measures what this interface is worth. Its cases differ only in what the decider knows. In cases A and B the decider knows only what air traffic control knows today from broadcast information – the aircraft's type, level, speed and route, from which case B assumes a nominal weight. In cases C and D it also has the items this interface carries. The difference between them is scored against the conflict of Section 3.1.4, at the level of the aircraft, the airline, the sector and the system.

4 Experiment

The experiment tests the airline-to-ATC interface proposed in Section 3.6, resolving around one conflict with and without the information carried by the proposed interface. Each theoretial resolution is scored at different system boundaries, including those of the aircraft, the airline, the sector and the system.

4.1 Scenario

The scenario used to explore the value of the proposed interface is a simple one – four aircraft in one invented en-route sector, shown as one snapshot depicted in Figure 10: a crossing pair at the same level and two aircraft in trail 2,000 ft below.

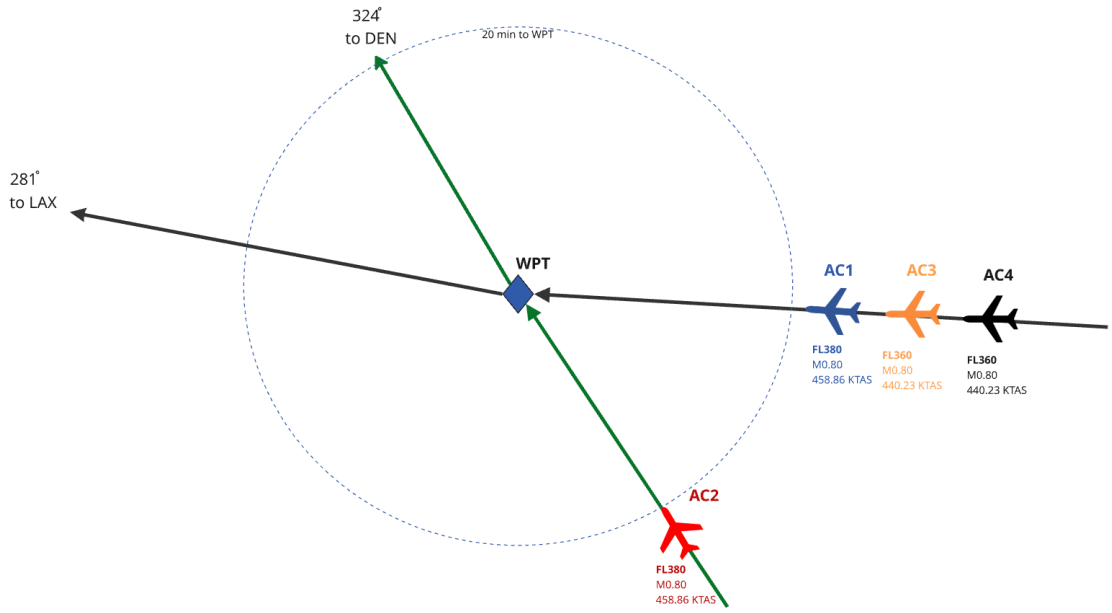


Figure 10: The scenario in plan view. AC1 and AC2 cross at the waypoint at FL380; AC3 and AC4 follow AC1's track at FL360.

The crossing pair is at FL380, which places the heavy aircraft is near its service ceiling (Figure 13). The two trailing aircraft are at FL360. The decision is made 12 minutes before AC1 and AC2 reach the waypoint. Figure 11 shows what follows when AC2 is moved down 4,000 ft: it descends for about 3.6 minutes, holds FL340 for 15 minutes while AC1, AC3 and AC4 pass, and climbs back for about 6.4 minutes. The 15-minute hold is about 5 minutes longer than the minimum that would let the three aircraft pass at minimum separation, which allows for delays in radio contact. Apart from this timeline, the scenario is a single snapshot that is scored, not simulated over time, as in the studies that informed it ([57], [35]).

Plan ref: to-do-list.md S11 (owner's note, 2026-10-11 print, page 11) — "Justify one scenario (time) and expand as possible." Say that one scenario is studied because of the time available in one term, name what a second scenario would add, and list the variations that are run on the one scenario (similar weights, the climb closed, a crossing pair at or below FL350).

4.2 Cases and Resolution Options

Plan ref: to-do-list.md S11 — More realistic comparison? Consider expanding to include a case that mimics EPP, which enables better understanding of the current trajectories and speed envelope (as evidenced by ETA windows). What else can be derived from EPP info in a split-second assessment? Also, should we throw "typical weight errors" into the tool and see their impact on the system's ability to make good decisions? How might that factor into the model?

Four cases are considered of varying levels of apriori information, and are summarized in Table 6. The decider's knowledge of the aircraft's perfomance and the relative expense of airline costs determines which resolution options it can offer.

Case A is the baseline, which is the current NAS practice. It has only the information that is currently available to the controller through current interfaces, which limits the options it can offer. [19] observed that controllers prefer to descend aircraft rather than climb them, and that they will not offer a climb if the aircraft is above FL350. The baseline case applies this rule, which limits

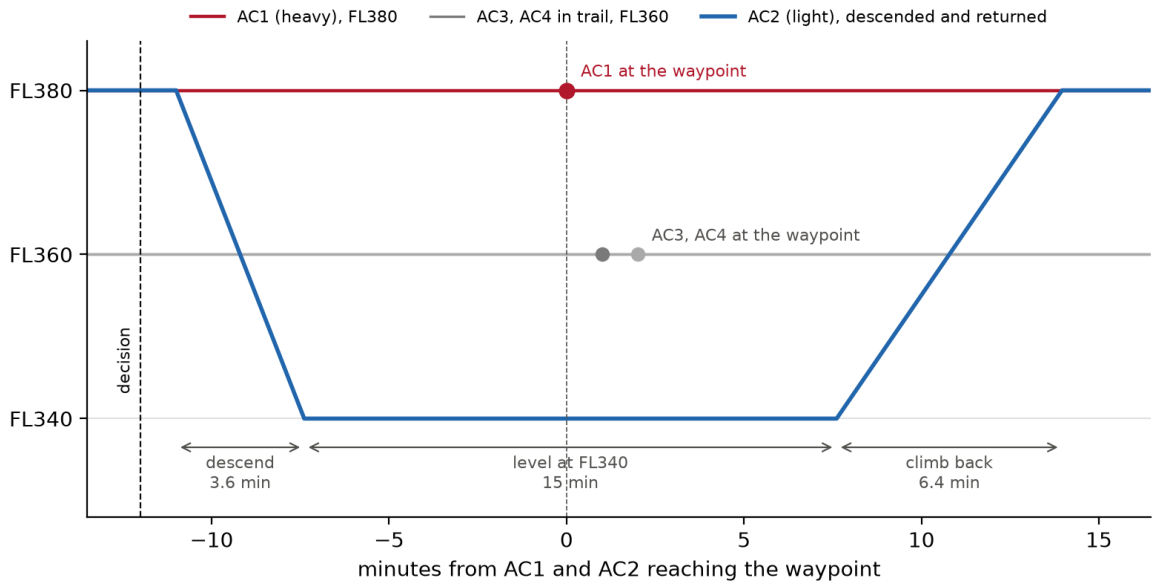


Figure 11: Level against time when AC2 is moved down 4,000 ft and back. The decision is made 12 minutes before AC1 and AC2 reach the waypoint and the descent starts a minute later; AC2 holds FL340 for 15 minutes while AC1, AC3 and AC4 pass, then climbs back. Descent and climb times are from OpenAP at AC2's mass.

Table 6: Conflict resolution options available to a controller with different levels of information about the aircraft.

Case	Decider	Information	Climb rule applies?
A	Controller, by rule	Type, level, speed, route	Yes
B	Controller, with decision aid	Nominal weight per type	Yes
C	Controller, with decision aid	Shared weight and cost index	No
D	Controller, with decision aid	Shared weight, cost index and delay margin	No

the options it can offer. [75] found that a single descent advisory was the most common resolution option.

Case B is an automated decision aid that has access to the nominal weight of each aircraft type, but not the actual weight or cost index. As such, the system does not have the information needed to guarantee that a climb is feasible. If an unachievable climb is proposed, the flight crew would have the obligation to reject it, leading to unacceptable delays in conflict resolution. As such, the system must also apply the climb rule in Case B.

Case C is an automated decision aid that has access to the actual weight and cost index of each aircraft. As such, it is not bound by the same limitations as Case B. It differs from Case D through the ability to account for the cost of delaying the time of arrival of the aircraft. The cost calculation os Case D incorporate a measure of the cost of delaying a flight with connections past its threshold, which is not available to Case C.

Table 7: Resolution options. A dash means no instruction. Full version with instruction counts in Appendix A.6.

Option	AC1	AC2	AC3	AC4
O1	Descend 2,000 ft	–	–	–
O2	–	Descend 2,000 ft	–	–
O3	Climb 2,000 ft	–	–	–
O4	–	Climb 2,000 ft	–	–
O5	Turn behind AC2	–	–	–
O6	–	Turn behind AC1	–	–
O7	Reduce speed	–	–	–
O8	Descend 4,000 ft	–	–	–
O9	Descend 2,000 ft	–	Slow 0.04 Mach	Slow 0.04 Mach
O10	–	Descend 4,000 ft	–	–

Plan ref: to-do-list.md S11 — Resolution options. State which aircraft descends and the tie-breaker (decided 2026-10-29: case A moves AC2, which has no in-trail traffic on its track), and whether the aircraft is cleared back up. For each option give the instructions it costs now, the follow-up it creates, and what the decider must know to use it. Cite [35] for the instruction accounting.

The conflict depicted in the scenario can be resolved by several options, which are listed in Table 7 and are available to all four cases, with the exception of options 3 and 4, which are not available for cases A and B for reasons described previously.

Options 1 and 2 are descents of the heavy and light aircraft, respectively. A descent is the most common resolution option observed in practice [75]. It is the safest option, as it is robust to heavy aircraft, which may be operating near their service ceiling with little room to maneuver. Cases A and B will be unable to discern a difference in cost between the two options due to limited knowledge.

Options 3 and 4 are climbs of the heavy and light aircraft, respectively, which are not available to cases A and B due to the climb rule. If A and B were able to consider climbs they would be similarly ignorant to the relative benefit and safety associated with the lighter aircraft climbing.

In Options 5 and 6, a lateral modification is assigned by the controller to achieve separation without the need for changes to flight level. Option 7 considers a speed reduction of the heavy aircraft, which is a less common resolution option, but which is practical and straight forward way to achieve separation if there is sufficient separation on either side of the pair. Aircraft 3 and 4 complicate its application to this scenario.

Option 8 is a 4,000 ft descent of the heavy aircraft, which a controller may deem necessary to simplify the separation with aircraft 3 and 4, which are in trail 2,000 ft below aircraft 1. Option 9 is a descent of the heavy aircraft to FL360 with AC3 and AC4 both slowed by 0.04 Mach (to Mach 0.76, about 440 KTAS) for 12 minutes, which opens a gap of more than 3 NM ahead of AC3 for AC1 to descend into; it is a less common resolution option and the only one that moves three aircraft. Option 10 is a 4,000 ft descent of the light aircraft, the mirror of option 8.

While not comprehensive, this set of options is representative of controller practices. [35] found that controllers relied on five lateral vectoring heuristics and eight vertical level-change heuristics to resolve potential conflicts between aircraft. [75] found that level-offs constituted the most frequent maneuver category, which are only applicable to aircraft ascending or descending and which represented 44%. Turns were the next most frequent, at 32%, followed by descents at 18%, and climbs at 5%. Turns occurred much less frequently than vertical trajectory changes,

accounting for 32% compared with 68%. These percentages reflect actions taken across all types of conflicts, rather than the cruise scenario alone.

Plan ref: to-do-list.md S11 — To Specify degree of turn, distance offset, and re-vectoring back on path, whether direct-to or otherwise. This will manifest in different costs.

4.3 Scoring

Each option is scored along several lines, to highlight the ways different actions lead to different costs and how those costs aggregate at varying system of interest boundaries, including:

- a single aircraft,
- all aircraft associated with a single airline,
- all aircraft in the sector,
- the airspace as a whole, including the cost of controller actions, airline schedule impacts, and environmental impacts.

Plan ref: to-do-list.md S11 — Review items. Still need to be checked against `simulation/experiment_scoring.py`. Also, need to finalize what costs will be included where, based on the models. Current content is a draft.

Individual aircraft costs are limited to time and fuel, and are equated through the use of a standard industry conversion with the cost index (CI). For aircraft i the cost of an option is its extra fuel plus its cost index times its extra time, in kilograms of fuel, the form used in [71]:

$$c_i = \Delta m_{f,i} + CI_i \, \Delta t_i, \tag{1}$$

where the extra fuel $\Delta m_{f,i}$ and the extra time Δt_i are the option minus the baseline of flying on undisturbed, and CI_i is the cost index of aircraft i in kilograms of fuel per minute. Boeing states the cost index as the cost of time in dollars per hour over the cost of fuel in cents per pound, on a scale of about 0 to 100, with fuel-saving operations below 15 and time-critical ones above 60; one unit is 100 lb of fuel per hour, or 0.756 kg/min. The scenario gives AC1 a cost index of 60 (45.4 kg/min), AC2 and AC3 10 (7.6 kg/min) and AC4 30 (22.7 kg/min).

An airline's cost is the sum over its aircraft plus a step cost when a flight misses a connection, plus CO2 converted to dollars. Fuel is priced at \$2.50 per kg (about \$7.60 per gallon, within the \$4.80 to \$8.47 per gallon seen for Jet-A with airport markups) and CO2 at \$15 per tonne, a CORSIA-style credit price.

The cost experienced by the sector include all individual aircraft costs as well as clearances issued, effort to devise them, and residual risk. The latter three costs are associated with the sector

Total airspace cost is that of the system of interest, and totals all costs experienced by airspace users and by the stakeholders of the system. As such, it includes the environmental cost associated with each action. This would also consider the balance of costs across airlines, if the scenario facilitated resolutions that lead to inequitable costs.

Plan ref: to-do-list.md S11 — Emissions. State the conversion from fuel to CO2 and its source ([50]). State how CO2 becomes dollars at the airline level and why that route was chosen (credit price, passenger value, or both; undecided), and why the system level keeps CO2 in kg.

4.3.1 Case Considerations

Case A applies its rule, one descent. Each automated case chooses, among the options it is allowed to offer, the option whose estimated total of Eq. 1 over all aircraft is lowest. Case B uses the type's nominal mass for every aircraft and values time at a typical cost index of 30 (22.7 kg/min). Cases C and D use each aircraft's actual mass and cost index. Case D also rejects an option that delays a flight with connections past its threshold. Whatever a case chooses is then scored at the actual masses, and that score is what the tables report.

Schedule reliability enters as a step cost, as described by [74], which occurs when a flight with connections is delayed past a given threshold. At the time threshold associated with missed connections, the flight incurs a cost K on top of Eq. 1. No sourced value of K is presently held, so the tables report whether the threshold is crossed, not a dollar figure. This may change in a future draft.

Carbon dioxide can be derived from the chemical formula of Jet-A fuel, as 3.16 kg per kilogram of fuel [50]. Instructions issued by controllers are counted for those required to resolve the initial conflict as well as those to clear the aircraft back to its original path, but are not presently converted to a USD cost.

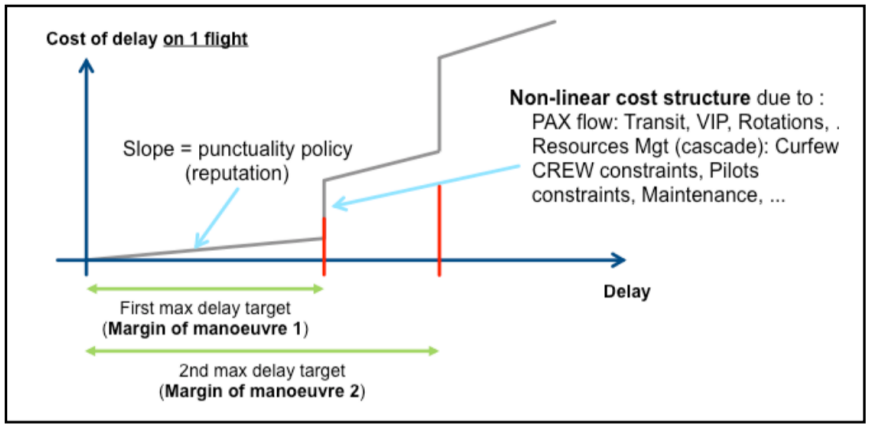


Figure 1: Typical cost-delay model profile per flight

Figure 12: Cost of delay for one flight, with steps where a margin is exceeded. Reproduced from [74], Figure 1.

4.4 Aircraft Model and Its Simplifications

Calculating a fuel and time cost involves defining a truth model to enable comparison of the resultant airspace performance observed as a result of resolving conflicts using the various optoins. The OpenAP (open aircraft performance model, TU Delft) is used to define that entitled information and supplies fuel flow, thrust and drag for the scenario aircraft, which are combined in the equations of motion to determine fuel and time deltas.

Each aircraft is a point mass in steady flight at a constant Mach number. Drag, maximum thrust and fuel flow are taken from OpenAP [90], which uses a two-term drag polar and, above 30,000 ft, the two-shaft turbofan thrust model of Bartel and Young [3].

Plan ref: to-do-list.md S11, S12 — OpenAP Review Read Sections 5 and 7 of the OpenAP paper (pp. 6 to 17) before submission. What the paper says about its own accuracy is in `evidence/literature-notes/summaries/4 - sun2020openap.md` and belongs in the discussion of the truth model's uncertainty (Section 5.2).

A descent is flown at OpenAP's typical constant-Mach descent rate for the type, about 1,100 ft/min. A climb is flown at maximum thrust, at the rate that thrust allows and no more than 1,500 ft/min. A level is available to an aircraft only if it would still have 150 ft/min of climb left there. Cases A and B do not know the mass and so cannot make that test; they offer a climb only at or below FL350, which the scenario does not explore. Cases C and D make it at the aircraft's actual mass. Figure 13 depicts a typical flight envelope, drawn assuming lift equals weight and using the OpenAP models.

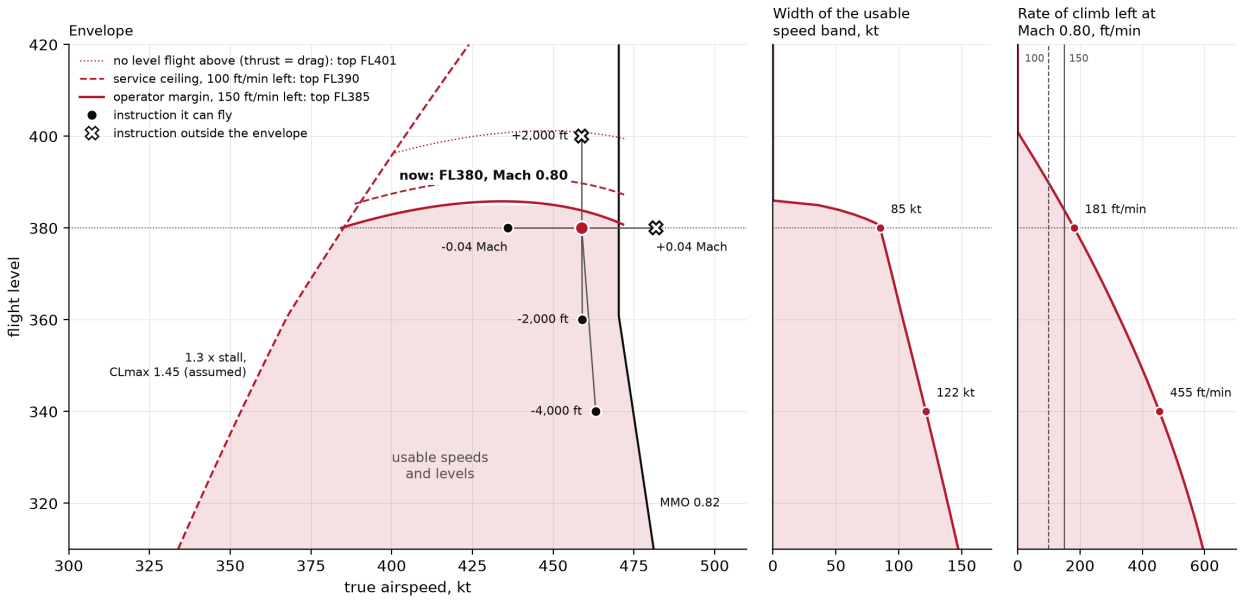


Figure 13: The heavy B737-9 (71.0 t) has little room left to change speed or level at FL380. Left: cruise envelope, with three lines of constant rate of climb left: 0 ft/min (thrust equals drag), 100 ft/min (the service ceiling, [22]) and 150 ft/min (the operator margin the scenario uses). Right: width of the usable speed band and rate of climb left at Mach 0.80, by flight level.

The low-speed boundary is 1.3 times the stall speed at an assumed clean maximum lift coefficient, and its high-speed boundary is the more restrictive of the maximum operating Mach number and the maximum operating speed (VMO). The aircraft's present level and Mach number are marked, and each altitude and speed instruction the scenario offers it is marked as inside or outside the region with 150 ft/min of climb left. Speed changs of +/- 0.04 Mach are presented, as these are typical simplifications employed by European controllers. The weight chosen for the figure shows the limitations associated with the speed simplifications, both in their inability to use the full speed change authority available to the aircraft and to forsee rejections that pilots will invevitably convey upon clearance issuance.

The model makes the following simplifications.

1. **Standard atmosphere, no wind, lift equal to weight.** Time differences come only from true airspeed and added track distance.

2. **No compressibility drag.** The drag used here has no Mach-number term, though its supported by OpenAP and has a non-trivial impact at Mach numbers above 0.75.

3. **The drag polar is the 737 MAX 8's.**

4. **The fuel curve is the package default.** OpenAP has no fuel model fitted to the 737 MAX 9 or MAX 8; it uses its default coefficients scaled by the LEAP-1B take-off fuel flow.

5. **AI-DRAFTED, owner to rewrite** — Fuel flow and altitude (D-041). Drafted 2026-11-07 at the owner's request.

   Fuel flow depends on thrust alone. OpenAP gives fuel flow as a function of the thrust asked of the engines, as a fraction of their sea-level maximum, with no altitude or Mach term. Every fuel difference between two levels here is therefore aerodynamic: in thinner air the lift coefficient rises toward its best lift-to-drag value ($\sqrt{C_{D0}/k} = 0.69$ for this polar) and drag falls. A real engine's fuel use for a given thrust also changes with altitude, which this model leaves out. Higher is not always better: at Mach 0.80 the least fuel per mile is at FL410 for AC1 and FL430 for AC2, where the lift coefficient is near 0.69, and above those levels drag rises again. Both lie above the highest level each aircraft can hold with 150 ft/min of climb left (FL380 and FL410), so in this scenario every level an aircraft can climb to is a better one.

6. **The climb margins.** The service ceiling is the level with 100 ft/min of climb left [22]. The 150 ft/min the experiment requires is commonly observed in practice.

7. **One climb rate per climb.** The rate is taken at the mid-altitude and held; the climb is not integrated. A climb is not checked against the time left before the crossing.

8. **Mass is constant.** Fuel burned during the manoeuvre does not reduce the mass.

9. **The low-speed limit rests on an assumed maximum lift coefficient** ($C_{L\,\max} = 1.45$) and is a stall margin, not a buffet boundary.

Plan ref: to-do-list.md S12 — Two more simplifications, found in the O4 hand-check walk-through (2026-11-07), for the owner to word into the list above. (1) *Flight-path angle.* OpenAP computes $\gamma = \arctan(v_s/V)$, treating true airspeed as the horizontal side; true airspeed lies along the path, so the exact angle is $\arcsin(v_s/V)$. For O4's climb (409 ft/min at 236 m/s) the two differ by 0.00002° and 0.001 kg of fuel: negligible, but a reader who checks by hand will find it. (2) *Round trip costed one way, in the default setting.* A descent's climb back is costed (O8, O10); a climb's descent back is not (O4). With `stay_at_new_level: if_better` (D-041), a climb whose new level is not better comes back down and the descent is costed; in this scenario every climb is to a better level, so the aircraft stays and nothing is left uncosted. Say which setting Table 8 uses (the default, `never`). Detail: `simulation/debug_o4_walkthrough.py`; journal 2026-11-07.

5 Results and Discussion (Preliminary)

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Results and Discussion (preliminary). For Interim Report #2 this section needs: (1) the architecture as it stands (context diagram, aircraft IBD, the three requirements tagged `improved` as the measured gap) — the figures are now in Section 3; (2) the two hand checks from `experiment-definition.md` §9 written up as preliminary results; (3) the first OpenAP numbers — Figure 14 and Table 8 are in place, the discussion is not; (4) one paragraph on what is not yet done.

5.1 Model Results

Plan ref: to-do-list.md S10, S13 (D-028) — Model results. With the architecture now presented in Section 3, report its measures here, against the measures named in Section 1.4: for example, how many stakeholder requirements trace to a kept objective, how many system requirements are allocated to a constituent system, and which requirements are tagged `improved` because today's NAS does not meet them. Without this, the systems engineering work is described but never assessed.

5.2 Experiment Results

Table 8: Outcome of each case at four levels, for the base scenario scored at the actual masses. Each cost is a difference from flying on undisturbed.

Case	Option	Largest cost to any one aircraft (kg fuel-equiv.)	Airline totals (USD)	Sector (calls / effort / risk)	System fuel, CO_2 (kg)
A	O10	40 (AC2)	X 0, Y 102, Z 0	1+1 / 1 / low	41, 130
B	O8	26 (AC1)	X 66, Y 0, Z 0	1+1 / 3 / low	32, 102
C	O4	12 (AC2)	X 0, Y 32, Z 0	1+1 / 3 / low	12, 39
D	O4	12 (AC2)	X 0, Y 32, Z 0	1+1 / 3 / low	12, 39

Table 9: For the option each case chose, whether the aircraft moved burns less fuel per nautical mile at its new level (OpenAP, at its actual mass, Mach 0.80), and what staying there would change. The cost in Table 8 covers the level change and the 15-minute hold either way. The last column is the rest of this scenario's flight after the hold; it is reported here and counted nowhere else.

Case	Option	Level change	Fuel per NM, planned / new (kg)	New level better?	Calls if it stays	After the hold, if it stays (this scenario's distance to go)
A	O10	AC2 FL380→FL340	4.37 / 4.61	no	–	returns
B	O8	AC1 FL380→FL340	4.81 / 4.97	no	–	returns
C	O4	AC2 FL380→FL400	4.37 / 4.29	yes	1+0	−39 kg, +0.0 min over 61 min
D	O4	AC2 FL380→FL400	4.37 / 4.29	yes	1+0	−39 kg, +0.0 min over 61 min

AI-DRAFTED, owner to rewrite — Staying at the new level (D-041). Drafted 2026-11-07 at the owner's request.

An aircraft cleared to a level where it burns less fuel has no reason to go back to its planned level once the conflict has cleared, and its crew would likely ask to stay. Table 9 makes that check for the option each case chose. In cases A and B the aircraft moved goes down 4,000 ft, to a level where it burns more fuel per mile, so it climbs back as Table 8 assumes. In cases C and D, AC2 goes up 2,000 ft to FL400, where it burns 4.29 kg per nautical mile against 4.37 at FL380, so it would likely stay. Staying changes two things. It removes the instruction to return, so cases C and D need one call instead of two (the run with `stay_at_new_level: if_better` gives 1+0, with the same fuel). And it leaves a benefit after the hold that Table 8 does not count. The experiment leaves that benefit out because it depends on how far the flight has left to go: counting it would tie the result to one route, and a flight close to its descent would see little of it. For this scenario's

AC2, 584 NM from Denver, the rest of the flight is about 76 minutes, 61 of them after the hold, and staying at FL400 would save about 39 kg more, assuming constant mass and cruise all the way to the destination. In the similar-weights variation, case C's climb (AC1 to FL400) gives the same answer: it would stay, and save about 59 kg over the 125 minutes after the hold. Table 8 is therefore conservative for cases C and D.

Plan ref: to-do-list.md S12 — Which option each case chose, and why. Walk Table 8 row by row. For case A, name the rule that produced the choice. For B, C and D, name the cost that decided it.

Plan ref: to-do-list.md S12 — Did the climb rule matter? With the crossing pair above FL350, cases A and B could not offer a climb and cases C and D could. Say whether case C used one, for which aircraft, and what it saved. If C did not climb either, say so: that is a result too.

Plan ref: to-do-list.md S12 — B minus A, and C minus B. The first is what automation is worth; the second is what the information exchange is worth. Compare the size of each with the prior figure of tens of gallons per flight ([19], p. 12) and say whether the difference is larger than the truth model's uncertainty.

Plan ref: to-do-list.md S9, S12 — Does the best option change with the level? Read across each row of Table 8. Name any option that is better for one aircraft or one airline and worse for the sector or the system, and who pays (the largest cost to any one aircraft). This is the sub-system versus system discussion.

Plan ref: to-do-list.md S12 — Airline X's two flights. Discuss the option that touches both of one airline's aircraft: what the airline would choose between its late flight and its early one, and that the controller cannot see that preference. Connect to the dispatcher priorities (schedule integrity and connections rated at or above fuel; source of that table still to confirm).

Plan ref: to-do-list.md S12 — Workload is three things. Discuss clearances issued, effort to devise, and residual risk separately. Say where a "more workload" option on one of them is "less" on another.

Plan ref: to-do-list.md S12 — When would the controller say no? (owner's note, second hand review, 5.2). Discuss whether refusals are more likely in morning rushes at hub airports or near holidays, using the sector-loading rule in [19] (p. 10). A hypothesis; needs a source.

Plan ref: to-do-list.md S12 — Is schedule the right priority? (owner's note, second hand review, 4.11). Contrast a carbon price, which makes fuel weigh more against time, with controllers' usual practice of giving schedule the priority.

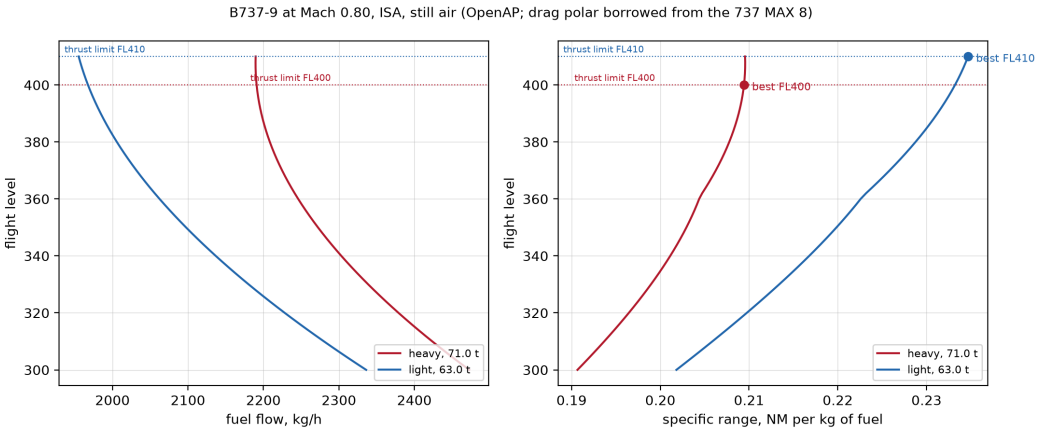


Figure 14: Fuel flow and specific range against altitude at Mach 0.80 for the heavy and light B737-9 (OpenAP).

Plan ref: to-do-list.md Track A — Figure slot: sequence diagram of case C's conflict resolution (controller, ATC decision support, airline operations center, FMS, crew), rendered from the model once drafted (turn 3).

Plan ref: to-do-list.md S12 — The case where nothing changes. Report the variation with similar weights and the climb closed, where case C picks the same option as case B, to show the comparison is not rigged.

Each column of Table 8 belongs to a part of the context diagram (Figure 4). The aircraft column belongs to the aircraft and its flight crew. The airline column belongs to the airline operations center within flight operations. The sector column belongs to air traffic control within airspace management. The system column belongs to no single part. It is a property of the NAS as a whole, which is the reason no constituent system computes it today and the reason the model is needed to state it.

The exchange that cases C and D add is one interface in the model: the airline operations center's performance share to air traffic control's decision support, carrying the aircraft's performance and the flight's delay margin. It satisfies system requirement SLR-INF-02, which is tagged as part of the improved architecture because today's NAS does not meet it. The interface is not drawn on Figure 4, which shows the baseline.

Reliability and resilience were set aside as objectives of the study (Section 3.1.3) because a single choice between resolution options does not change them. Across many conflicts, more efficient resolution would serve them as a side benefit: a resolution that costs less time and fewer instructions leaves each flight closer to its plan and the sector with more margin for the next disruption. The experiment does not measure this.

5.3 Emissions and Who Pays

Plan ref: to-do-list.md S9, S12 — Airline level: put CO2 into dollars. Take the fuel difference between cases from Table 8, convert to kg of CO2, then to dollars. State the route used and its source. Candidates: (1) the price of a carbon credit or allowance, if a scheme applies to the flight; (2) passenger value, meaning revenue gained or lost because some passengers choose the flight listed with lower emissions. Make the point that fuel is already counted once as fuel cost and that pricing its CO2 counts the same kilograms again, on purpose.

Plan ref: to-do-list.md S9, S12 — Passenger value. Discuss how passengers see a flight's emissions when booking, and whether that reaches the airline as demand. The author's own practice of choosing lower-emission listings is the motivation, not the evidence: cite a study. Note the gap between one decision in one sector and the label a passenger saw at booking; the link is the airline's average over many flights.

Plan ref: to-do-list.md S12 — Scale it honestly. One conflict in one sector is a small number. If a per-year or fleet-wide figure is offered, state the multiplier and its source (for example, how often level crossing conflicts occur), and label it illustrative.

Plan ref: to-do-list.md S9 — Who pays. For the option the system prefers, name the aircraft or airline that is worse off and by how much. This is the equity discussion.

6 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its experiment. Translate the conflict analysis and the experiment's results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO2, instructions issued) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to the objectives whose costs are externalized onto another stakeholder, Section 3.1.4, and to Section 5.3).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template's sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a "before" baseline in dollars, so

a literal dollar-savings figure may not be available or meaningful. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

7 Conclusions and Recommendations

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., "reduces network-level delay propagation by an illustrative X% under the demonstrated scenario" rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the experiment (Sections 2.2 and 4.4), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

Plan ref: to-do-list.md S14, S15 — Recommendations. Restate the main findings from Section 5 and provide recommendations for moving forward. Include the alternatives considered (alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9's catalog).

References

[1] Airbus Customer Services. *Flight Operations Briefing Notes: Approach Techniques: Aircraft Energy Management during Approach*. Tech. rep. FLT_OPS – APPR – SEQ03 – REV02. Airbus, Oct. 2005. URL: https://skybrary.aero/sites/default/files/bookshelf/166.pdf.

[2] Magnus Axholt and Stephen Peterson. "Modelling Traffic Scenarios for Realistic Air Traffic Control Environment Testing". MA thesis. Linköping University, 2004.

[3] Matthias Bartel and Trevor M. Young. "Simplified Thrust and Fuel Consumption Models for Modern Two-Shaft Turbofan Engines". In: *Journal of Aircraft* 45.4 (2008). Not held and not read. Authors, title, journal, volume, issue, pages and DOI confirmed 2026-10-11 against the Crossref record for the DOI (DOI supplied by the owner)., pp. 1450–1456. DOI: 10.2514/1.35589.

[4] Jason E. Bartolomei et al. "Engineering Systems Multiple-Domain Matrix: An Organizing Framework for Modeling Large-Scale Complex Systems". In: *Systems Engineering* 15.1 (2012), pp. 41–61. DOI: 10.1002/sys.20193.

[5] Katherine A. Berry and John J. Pace. "Examining the Actors and Functions of an Airline Operations Center". In: *Proceedings of the Human Factors and Ergonomics Society 55th Annual Meeting*. Vol. 55. 1. 2011, pp. 1412–1416. DOI: 10.1177/1071181311551294.

[6] Karl D. Bilimoria, Mike M. Paglione, and Hilda Q. Lee. "Performance Analysis of a Conflict Probe Utilizing Only State Vector Information". In: *24th International Congress of the Aeronautical Sciences (ICAS)*. Yokohama, Japan, 2004. URL: https://www.icas.org/icas_archive/ICAS2004/PAPERS/216.PDF.

[7] Katie Bolczak et al. *FAA Data Standards Initiative: Systems Engineering Base for Air Traffic Modernization*. Tech. rep. The MITRE Corporation, 2001.

[8] Soufiane Bouarfa, Jasper Müller, and Henk A. P. Blom. "Evaluation of a Multi-Agent System Approach to Airline Disruption Management". In: *Journal of Air Transport Management* 71 (2018), pp. 108–118. DOI: 10.1016/j.jairtraman.2018.05.009.

[9] Chris Brady. *Rules of Thumb*. The Boeing 737 Technical Site (web page). 2026. URL: http://www.b737.org.uk/rulesofthumb.htm.

[10] Ian Britchford. "737 MAX 8 & 9 fuel burn and operating performance". In: *Aircraft Commerce* 137 (Aug. 2021), pp. 18–26. URL: https://www.aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/General%20Articles/2021/137_FLTOPS.pdf.

[11] António Jesus Monteiro de Castro. "A Distributed Approach to Integrated and Dynamic Disruption Management in Airline Operations Control". PhD thesis. Porto, Portugal: Faculdade de Engenharia, Universidade do Porto (FEUP), 2013.

[12] Mohammad Chami et al. "Towards Solving MBSE Adoption Challenges: The D3 MBSE Adoption Toolbox". In: *28th Annual INCOSE International Symposium*. Washington, DC, 2018.

[13] Gano Chatterji. *Introduction to Air Traffic Management*. Video and transcript, Aviation Data Science Seminar (virtual). NASA Technical Reports Server (NTRS), Document ID 20205001251. NASA Ames Research Center; Universities Space Research Association, Apr. 2020. URL: https://ntrs.nasa.gov/citations/20205001251 (visited on 09/20/2026).

[14] Michael D. D. Clarke. "Irregular Airline Operations: A Review of the State-of-the-Practice in Airline Operations Control Centers". In: *Journal of Air Transport Management* 4 (1998), pp. 67–76.

[15] Richard A. Coppenbarger. "En Route Climb Trajectory Prediction Enhancement Using Airline Flight-Planning Information". In: *AIAA Guidance, Navigation, and Control Conference and Exhibit*. AIAA-99-4147. 1999, pp. 1077–1087. DOI: 10.2514/6.1999-4147.

[16] Daniel A. DeLaurentis. "Understanding Transportation as a System-of-Systems Design Problem". In: *43rd AIAA Aerospace Sciences Meeting and Exhibit*. Reno, Nevada, 2005. DOI: 10.2514/6.2005-123.

[17] Daniel A. Delaurentis and Donald N. Fry. "Understanding the Implications for Airports of Distributed Air Transportation Using a System-of-Systems Approach". In: *Transportation Planning and Technology* 31.1 (2008), pp. 69–92. DOI: 10.1080/03081060701835738.

[18] Ahmed E. E. Eltoukhy, Felix T. S. Chan, and S. H. Chung. "Airline Schedule Planning: A Review and Future Directions". In: *Industrial Management & Data Systems* 117.6 (2017), pp. 1201–1243. DOI: 10.1108/IMDS-09-2016-0358.

[19] Engility Corporation. *Traffic Aware Strategic Aircrew Requests (TASAR): Annualized TASAR Benefits for Alaska Airlines Operations*. Contractor report NNL12AA06C, deliverable 41A. Corporate author, as the document states: "prepared by Engility Corporation ... under Contract No. NNL12AA06C with NASA Langley Research Center" (Preface, p. 2); NASA Technical Monitor: David Wing (title page). No personal author is named. A web search summary of the NASA record for a later published version (NASA/CR-2015-218787, NTRS 20150017048) names Jeffrey Henderson; not confirmed and not used here. Engility Corporation, Billerica, MA, for NASA Langley Research Center, Sept. 2014. URL: https://ntrs.nasa.gov/api/citations/20140012787/downloads/20140012787.pdf.

[20] EUROCONTROL. *EUROCONTROL Specification for Airport Collaborative Decision Making (A-CDM)*. Tech. rep. Brussels, Belgium: EUROCONTROL, 2025.

[21] EUROCONTROL. *Aircraft Performance Database: BOEING 737 MAX 9 (B39M)*. Web page. 2026. URL: https://learningzone.eurocontrol.int/ilp/customs/ATCPFDB/details.aspx?ICAO=B39M.

[22] Federal Aviation Administration. *Aircraft Operations at Altitudes Above 25,000 Feet Mean Sea Level or Mach Numbers Greater Than .75*. Advisory Circular AC 61-107B, Change 1. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, Sept. 2015. URL: https://www.faa.gov/regulations_policies/advisory_circulars.

[23] Federal Aviation Administration. *Instrument Procedures Handbook, Chapter 3: Arrivals*. Tech. rep. FAA-H-8083-16B. U.S. Department of Transportation, Federal Aviation Administration, 2017. URL: https://www.faa.gov/sites/faa.gov/files/regulations_policies/handbooks_manuals/aviation/instrument_procedures_handbook/FAA-H-8083-16B_Chapter_3.pdf.

[24] Federal Aviation Administration. *Flight Information Exchange Model (FIXM) US Extension v4.4.0: Logical Model Diagrams*. Tech. rep. Federal Aviation Administration / FIXM Program, 2024.

[25] Federal Aviation Administration. *Aeronautical Information Manual: Official Guide to Basic Flight Information and ATC Procedures*. Aeronautical Information Manual. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, Feb. 2025. URL: https://www.faa.gov/air_traffic/publications.

[26] Federal Aviation Administration. *Air Traffic by the Numbers*. Fact book 2025-AJO-094. 9th edition, data through fiscal year 2024. Cover dated June 2025; the introduction is signed May 2025 by the System Events and Analysis Group (AJR-G3). URL found by search 2026-11-05; the file name on disk matches it. Washington, DC: FAA Air Traffic Organization, System Operations Services, Office of Performance Analysis, June 2025. URL: https://www.faa.gov/air_traffic/by_the_numbers/air-traffic-by-the-numbers-FY2024.pdf.

[27] Federal Aviation Administration. *Air Traffic Control*. Order JO 7110.65BB. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[28] Federal Aviation Administration. *FAA Services Hierarchy*. One-page slide (PDF), FAA NAS Enterprise Architecture. Year is the PDF creation date (2025-02-24, PowerPoint export; PDF author field "ANG-B2"); no version, title-page date or byline appears on the slide. Unverified: what ANG-B2 stands for, and whether this is the current issue of the hierarchy on the FAA NAS EA site (faa.gov/about/office_org/headquarters_offices/ang/r2r) – check before citing as current. 2025.

[29] Federal Aviation Administration. *Facility Operation and Administration*. Order JO 7210.3EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[30] Federal Aviation Administration. *Flight Services*. Order JO 7110.10EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[31] Federal Aviation Administration. *NAS Enterprise Architecture: Infrastructure Roadmaps*. Tech. rep. Federal Aviation Administration, 2025.

[32] Federal Aviation Administration. *Airport Compliance Manual, Chapter 12: Review of Aeronautical Lease Agreements*. Order 5190.6C. URL not recorded when the chapter was downloaded on 2026-09-28; add it from the FAA Order 5190.6C page before citing. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026.

[33] Federal Aviation Administration. *Airport Compliance Manual, Chapter 9: Unjust Discrimination between Aeronautical Users*. Order 5190.6C. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026. URL: https://www.faa.gov/airports/resources/publications/orders/compliance_5190_6/order-5190-6c-compliance-chapter-09 (visited on 09/28/2026).

[34] Michael Finke et al. *D2.1: Current TBO Concepts and Derivation of the Green Air Traffic Management Concepts*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2020.

[35] Selina Fothergill and Andrew Neal. "Conflict-Resolution Heuristics for En Route Air Traffic Management". In: *Proceedings of the Human Factors and Ergonomics Society 57th Annual Meeting*. 2013, pp. 71–75. DOI: 10.1177/1541931213571018.

[36] Sanford Friedenthal and Joseph Wolfrom. *Modeling with SysML*. Tutorial slides, INCOSE International Symposium 2010, Chicago, IL. 148 slides. Affiliations on slide 1: Lockheed Martin (Friedenthal), JHU/APL (Wolfrom). The PDF's metadata title ("Briefing at introductory meeting with AFIT students and Advisor") is left over from a slide template; the title is from slide 1. URL is the JHU/APL copy found by search 2026-11-05 under the same file name; not re-downloaded to compare. July 2010. URL: https://jhuapl.edu/sites/default/files/2023-03/ModelingwithSysMLTutorial.pdf.

[37] GE Aerospace. *GE Aerospace Joins MBSE Ecosystem in Michigan Aerospace*. University of Michigan College of Engineering, Aerospace Engineering news. May 2023. URL: https://aero.engin.umich.edu/2023/05/11/ge-aerospace-joins-mbse-ecosystem-in-michigan-aerospace/ (visited on 09/14/2026).

[38] L. K. Hassan, B. F. Santos, and J. Vink. "Airline Disruption Management: A Literature Review and Practical Challenges". In: *Computers & Operations Research* 127 (2021), p. 105137. DOI: 10.1016/j.cor.2020.105137.

[39] Terry R. Hill et al. "NASA's Use of MBSE and SysML Modeling to Architect the Future of Human Exploration". In: *34th Annual INCOSE International Symposium*. Dublin, Ireland, 2024.

[40] Eric C. Honour. "Systems Engineering Return on Investment". In: *INCOSE International Symposium*. Vol. 20. 1. Chicago, IL, 2010, pp. 1422–1439. DOI: 10.1002/j.2334-5837.2010.tb01150.x.

[41] Eric C. Honour. "Sizing Systems Engineering Activities to Optimize Return on Investment". In: *INCOSE International Symposium*. Vol. 21. 1. Denver, CO, 2011, pp. 1233–1245. DOI: 10.1002/j.2334-5837.2011.tb01281.x.

[42] Eric C. Honour. "Systems Engineering Return on Investment". PhD thesis. Adelaide, Australia: Defence, Systems Institute, School of Electrical, and Information Engineering, University of South Australia, 2013. URL: https://segoldmine.ppi-int.com/node/72248 (visited on 09/19/2026).

[43] Niamat Ullah Ibne Hossain et al. "Modeling and Analysis of Unmanned Aerial Vehicle System Leveraging Systems Modeling Language (SysML)". In: *Systems* 10.6 (2022), p. 264. DOI: 10.3390/systems10060264.

[44] Haoliang Hu et al. *D2.2: Report about Modelling of Operational and System Architecture*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[45] Haoliang Hu et al. *D5.1: Report on ATM Avionic System Architecture Development*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[46] Yuzhen Hu et al. "Review of Optimization Problems, Models and Methods for Airline Disruption Management from 2010 to 2024". In: *Digital Transportation and Safety* 3.4 (2024), pp. 246–263. DOI: 10.48130/dts-0024-0022.

[47] INCOSE. *Guide to Writing Requirements*. Tech. rep. INCOSE-TP-2010-006-04. Version 4. International Council on Systems Engineering, 2023.

[48] INCOSE. *INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*. 5th. Wiley, 2023. ISBN: 978-1-119-81429-0. URL: https://www.incose.org/about-systems-engineering/system-and-se-definition/systems-engineering-definition (visited on 09/19/2026).

[49] International Civil Aviation Organization. *Global Air Traffic Management Operational Concept*. Doc 9854, AN/458. Montreal, Canada: International Civil Aviation Organization, 2005.

[50] International Civil Aviation Organization. *ICAO Carbon Emissions Calculator Methodology*. Methodology document Version 13.1. International Civil Aviation Organization, Aug. 2024. URL: https://www.icao.int/sites/default/files/environmental-protection/ENVtools/SuportingMaterials/ICEC_Methodology_Passagers_v13_Final.pdf.

[51] International Civil Aviation Organization. *ICAO Aircraft Engine Emissions Databank*. Spreadsheet, hosted by the European Union Aviation Safety Agency. Issue 32. Mar. 2026. URL: https://www.easa.europa.eu/easa-and-you/environment/icao-aircraft-engine-emissions-databank.

[52] ISO/IEC/IEEE. *ISO/IEC/IEEE 15288:2023 Systems and software engineering — System life cycle processes*. Tech. rep. Geneva: International Organization for Standardization, 2023.

[53] Swapnil Sarjerao Jagtap et al. "Model-Based Systems Engineering Approach for a Systematic Design of Aircraft Engine Inlet". In: *AIAA SciTech Forum 2025*. 2025. DOI: 10.2514/6.2025-1410.

[54] Priti Jain. "Personal Knowledge Management: The Foundation of Organisational Knowledge Management". In: *South African Journal of Libraries and Information Science* 77.1 (2011), pp. 1–14. DOI: 10.7553/77-1-62.

[55] Abdessamad Jarrar and Youssef Balouki. "Formal Modeling of a Complex Adaptive Air Traffic Control System". In: *Complex Adaptive Systems Modeling* 6.1 (2018), p. 6. DOI: 10.1186/s40294-018-0056-4.

[56] Heiko Kern, Axel Hummel, and Stefan Kühne. "Towards a Comparative Analysis of Meta-Metamodels". In: *Proceedings of the Compilation of the Co-located Workshops on DSM'11, TMC'11, AGERE! 2011, AOOPES'11, NEAT'11, & VMIL'11 (SPLASH 2011 Workshops)*. ACM, 2011, pp. 7–12. DOI: 10.1145/2095050.2095053.

[57] Barry Kirwan and Mary Flynn. "Identification of Air Traffic Controller Conflict Resolution Strategies for the CORA (Conflict Resolution Assistant) Project". In: *4th USA/Europe Air Traffic Management R&D Seminar*. Santa Fe, NM, Dec. 2001.

[58] Konstantia Kontodimou et al. "Proactive Aircraft Turnaround Buffer Optimization Integrating Machine Learning and Scenario Analysis". In: *Journal of the Air Transport Research Society* 6 (2026), p. 100117. DOI: 10.1016/j.jatrs.2026.100117.

[59] Deepak Kulkarni. *Impact of Slot Substitution on CTOP Performance*. Technical Memorandum NASA/TM-2019-220158. Moffett Field, CA: NASA Ames Research Center, Mar. 2019.

[60] Tianning Liu et al. "Model-Based Systems Engineering Supporting Architecture Modeling of Air Traffic Management System and Model Verifying Based on SMT". In: *Knowledge Science, Engineering and Management (KSEM 2024)*. Vol. 2269. Communications in Computer and Information Science. Springer, 2025, pp. 183–197. DOI: 10.1007/978-981-96-0178-3_13.

[61] Manoj Lohatepanont and Cynthia Barnhart. "Airline Schedule Planning: Integrated Models and Algorithms for Schedule Design and Fleet Assignment". In: *Transportation Science* 38.1 (2004), pp. 19–32. DOI: 10.1287/trsc.1030.0026.

[62] Jing Lu et al. "Harnessing Digital Twin Technology for Enhanced Aircraft Turnaround Efficiency". In: *Transport Economics and Management* 3 (2025), pp. 334–345. DOI: 10.1016/j.team.2025.09.001.

[63] Jinzhi Lu et al. "Design Ontology Supporting Model-based Systems-engineering Formalisms". In: *arXiv preprint* (2020).

[64] Jinzhi Lu et al. "Semantic Model-based Systems Engineering based on KARMA: A Research and Practice Roadmap 2025". In: *32nd Annual INCOSE International Symposium*. Detroit, MI, 2022.

[65] Christopher A. Lupp et al. "A Theoretical Foundation and Practical Demonstration of Integrating MDO, MBSE and the Digital Thread". In: *AIAA SciTech Forum*. Air Force Research Laboratory / University of Dayton Research Institute. 2026.

[66] Apoorv Maheshwari, C. Robert Kenley, and Daniel A. DeLaurentis. "Creating Executable Agent-Based Models Using SysML". In: *25th Annual INCOSE International Symposium (IS2015)*. Bellevue, WA, 2015.

[67] N. A. M. Maiden et al. "Model-Driven Requirements Engineering: Synchronising Models in an Air Traffic Management Case Study". In: *16th International Conference on Advanced Information Systems Engineering (CAiSE 2004)*. Vol. 3084. Lecture Notes in Computer Science. Riga, Latvia: Springer, 2004, pp. 368–383.

[68] Tom McDermott et al. "AI4SE and SE4AI: A Research Roadmap". In: *INSIGHT* 23.1 (2020), pp. 8–14. DOI: 10.1002/inst.12278.

[69] Faida Mhenni et al. "Flight Control System Modeling with SysML to Support Validation, Qualification and Certification". In: *13th IFAC and IEEE Conference on Programmable Devices and Embedded Systems*. Vol. 49-3. IFAC-PapersOnLine. Elsevier, 2016, pp. 453–458. DOI: 10.1016/j.ifacol.2016.07.076.

[70] Yaniv Mordecai. "Conceptual Modeling of Cyber-Physical Gaps in Air Traffic Control". In: *Complex Adaptive Systems Conference with Theme: Cyber Physical Systems and Deep Learning (CAS 2018)*. Vol. 140. Procedia Computer Science. Elsevier, 2018, pp. 21–28. DOI: 10.1016/j.procs.2018.10.288.

[71] Ryota Mori. "Aircraft Mass Estimation Using Cruise Flight Profile". In: *33rd Congress of the International Council of the Aeronautical Sciences (ICAS)*. Stockholm, Sweden, 2022. URL: https://www.icas.org/icas_archive/ICAS2022/data/papers/ICAS2022_0105_paper.pdf.

[72] Sai V. Mudumba et al. "Model-Based Systems Engineering Approach for Simulating Urban Air Mobility Maturity Level 5 (UML-5) Operations". In: *AIAA AVIATION Forum*. 2022. DOI: 10.2514/6.2022-4075.

[73] Pamela Munro and Richard Mogford. "Managing Variability: A Cognitive Ethnography of the Work of Airline Dispatchers". In: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. Vol. 62. 1. 2018, pp. 182–186. DOI: 10.1177/1541931218621043.

[74] Nadine Pilon, Laurent Guichard, and Katherine Cliff. "Reducing Impact of Delays using Airspace User-Driven Flight Prioritisation: User Driven Prioritisation Process Validation Simulation and Results". In: *9th SESAR Innovation Days*. Paper 39. ISSN 0770-1268. Dec. 2019. URL: https://sesarju.eu/sites/default/files/documents/sid/2019/papers/SIDs_2019_paper_39-1.pdf.

[75] Esa M. Rantanen and Christopher D. Wickens. "Conflict Resolution Maneuvers in Air Traffic Control: Investigation of Operational Data". In: *The International Journal of Aviation Psychology* 22.3 (2012), pp. 266–281. DOI: 10.1080/10508414.2012.691048.

[76] Bill Roberson. "Fuel Conservation Strategies: Cost Index Explained". In: *Boeing AERO* QTR_02 (2007), pp. 26–28. URL: https://skybrary.aero/sites/default/files/bookshelf/1956.pdf.

[77] Ítalo Romani de Oliveira et al. "A Predictive Services Architecture for Efficient Airspace Operations". In: *2026 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. IEEE. 2026. DOI: 10.1109/ICNS69853.2026.11570268.

[78] Ahmed R. Sadik et al. "Urban Air Mobility as a System of Systems: An LLM-Enhanced Holonic Approach". In: *20th Annual System of Systems Engineering Conference (SOSE)*. IEEE. 2025.

[79] Mateus Santana et al. "The Aircraft Recovery Problem: A Systematic Literature Review". In: *EURO Journal on Transportation and Logistics* 12 (2023), p. 100117. DOI: 10.1016/j.ejtl.2023.100117.

[80] Michael Schmidt. "A Review of Aircraft Turnaround Operations and Simulations". In: *Progress in Aerospace Sciences* 92 (2017), pp. 25–38. DOI: 10.1016/j.paerosci.2017.05.002.

[81] Charles A. Schultz, David Thipphavong, and Heinz Erzberger. "Adaptive Trajectory Prediction Algorithm for Climbing Flights". In: *AIAA Guidance, Navigation, and Control Conference*. AIAA-2012-4931. Venue and paper number unconfirmed from the PDF; DOI not found. 2012. URL: https://ntrs.nasa.gov/api/citations/20140004896/downloads/20140004896.pdf.

[82] Florian Schummer and Maximillian Hyba. "An Approach for System Analysis with Model-based Systems Engineering and Graph Data Engineering". In: *Data-Centric Engineering* 3 (2022), e33. DOI: 10.1017/dce.2022.33.

[83] Thomas L. Seamster, Eric Chevalley, and Barbara G. Kanki. *Collaborative Systems Assessment: Flightdeck, Air Traffic Control, Flight Operations Center and Automation*. Draft Final Report, FAA task 05-04 / AJP61SSP-0050. NASA Ames Research Center, Aug. 2011.

[84] SESAR Joint Undertaking. *SESAR Solutions Catalogue 2019*. Tech. rep. Catalogue number MG-07-18-083-EN-N. Luxembourg: Publications Office of the European Union, 2019. DOI: 10.2829/998701.

[85] SESAR Joint Undertaking. *European ATM Master Plan: Making Europe the Most Efficient and Environmentally Friendly Sky to Fly in the World*. Tech. rep. SESAR Joint Undertaking, 2025.

[86] SESAR Joint Undertaking. *SESAR eATM ATM Capability Model (Extracted Capability Catalogue)*. 2025.

[87] Kapil Sheth and Sebastian Gutierrez-Nolasco. "Analysis of Factors for Incorporating User Preferences in Air Traffic Management: A System Perspective". In: *27th International Congress of the Aeronautical Sciences (ICAS 2010)*. Nice, France, 2010.

[88] Kapil S. Sheth et al. "Simulations of Credits Concept with User Input for Collaborative Air Traffic Management". In: *AIAA Guidance, Navigation, and Control Conference*. AIAA 2010-8079. Toronto, Ontario, Canada, Aug. 2010. DOI: 10.2514/6.2010-8079.

[89] Sonali Sinha Roy et al. "From Ontology to System Architecture: An MBSE Approach Toward the Realization of Urban Air Mobility". In: *34th Congress of the International Council of the Aeronautical Sciences (ICAS)*. Florence, Italy, 2024.

[90] Junzi Sun, Jacco M. Hoekstra, and Joost Ellerbroek. "OpenAP: An Open-Source Aircraft Performance Model for Air Transportation Studies and Simulations". In: *Aerospace* 7.8 (2020), p. 104. DOI: 10.3390/aerospace7080104. URL: https://openap.dev (visited on 10/10/2026).

[91] Serkan Turhan, Fatma Gzara, and Samir Elhedhli. "Workload Balancing for Flight Dispatchers". In: *Computers & Operations Research* 186 (2026), p. 107303. DOI: 10.1016/j.cor.2025.107303.

[92] Sabine Wilke, Arnab Majumdar, and Washington Y. Ochieng. "Airport Surface Operations: A Holistic Framework for Operations Modeling and Risk Management". In: *Safety Science* 63 (2014), pp. 18–33. DOI: 10.1016/j.ssci.2013.10.015.

[93] Tim Wittenborg et al. *Knowledge-Based Aerospace Engineering: A Systematic Literature Review*. 2025.

[94] Yifan Xu, Sebastian Wandelt, and Xiaoqian Sun. "Airline Scheduling Optimization: Literature Review and a Discussion of Modelling Methodologies". In: *Intelligent Transportation Infrastructure* 2 (2024), liad026. DOI: 10.1093/itinfr/liad026.

[95] Anzhuo Yao et al. "System-of-Systems Safety for Low-Altitude Aviation Transportation". In: *Reliability Engineering and System Safety* 273 (2026), p. 112276. DOI: 10.1016/j.ress.2026.112276.

[96] Haytham Younus et al. "AI- and Ontology-Based Enhancements to FMEA for Advanced Systems Engineering: Current Developments and Future Directions". In: *Applied Sciences* 16.5 (2026), p. 2464. DOI: 10.3390/app16052464.

A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 3.4). The phase activity diagrams are placed below (Fly Enroute is in the body, Figure 6); add the text that introduces them and cite each one from it.

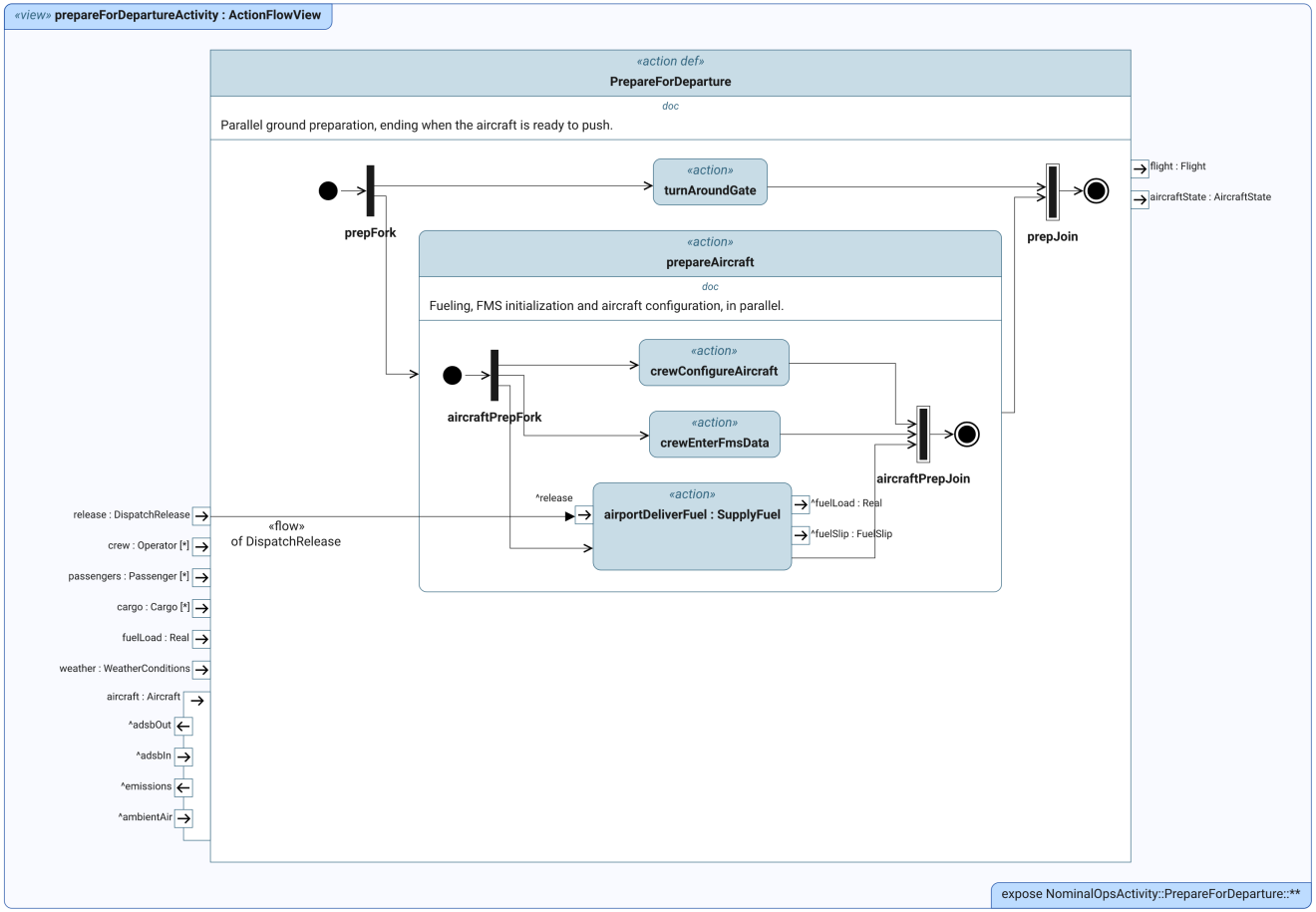


Figure 15: Nominal operations, Prepare for Departure: parallel fueling, FMS entry, configuration and gate turnaround. View `prepareForDepartureActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-19.

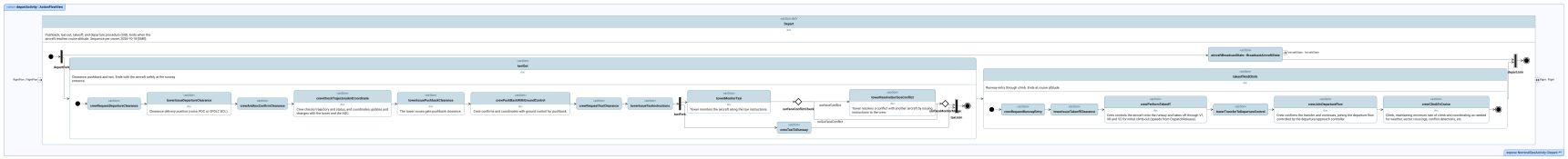


Figure 16: Nominal operations, Depart: clearance, pushback, taxi-out, takeoff and climb. View `departActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-19.



Figure 17: Nominal operations, Arrive: arrival planning, descent, approach, landing and taxi-in. View `arriveActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-11-05.

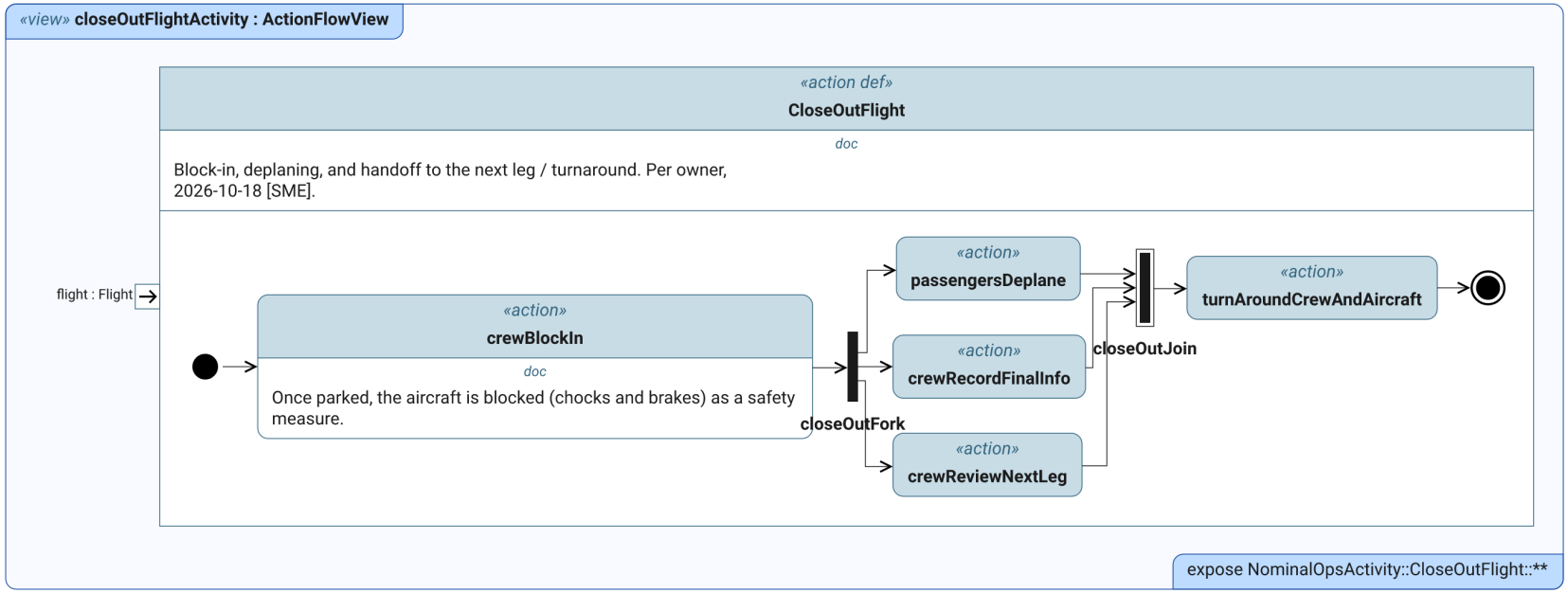


Figure 18: Nominal operations, Close Out Flight: block-in, deplaning and turnaround. View `closeOutFlightActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.

A.2 Full Stakeholder Register

Plan ref: to-do-list.md S7 — Introduce the two tables below and cite each from the body: the stakeholder assessment (Table 10) and the trace from each stakeholder objective to the requirements (Table 11).

Table 10: Stakeholder assessment: all 48 candidate stakeholders, sorted by decision score (`capstone_stakeholder_assessment.xlsx`). Score is standalone importance (high 3, medium 2, low 1) plus trajectory-decision impact (yes 2, indirect 1, no 0); the workbook's other columns (MOPs, MOEs, rationale) are not shown.

PESTLE domain	Stakeholder	Stated interest	Trajectory impact	Importance	Score	Recommended treatment
Economic	Adjacent Airspace Users	Throughput and reliability	Yes	High	5	Standalone
Technical	Air traffic controllers	Predictable aircraft behavior	Yes	High	5	Standalone
Economic	Airlines	Profitability	Yes	High	5	Standalone
Economic	Airports	Revenue and capacity	Yes	High	5	Standalone
Economic	Cargo operators	Throughput and reliability	Yes	High	5	Standalone
Technical	Dispatchers	Operational optimization	Yes	High	5	Standalone
Political	Militaries	Navigable airspace & resources with priority	Yes	High	5	Standalone
Economic	Engine manufacturers	Product performance and aftermarket revenue	Indirect	High	4	Consider / Consolidate
Social	Flight crews	Workload and quality of life	Indirect	High	4	Consider / Consolidate
Social	Local communities	Noise and environmental impacts	Indirect	High	4	Consider / Consolidate
Environmental	Local communities	Reduced noise	Indirect	High	4	Consider / Consolidate
Social	Business travelers	Schedule reliability	Indirect	Medium	3	Consider / Consolidate
Environmental	Environmental advocacy groups	Reduced emissions	Indirect	Medium	3	Consider / Consolidate
Political	Federal Aviation Administration	Safe and efficient airspace	No	High	3	Consider / Consolidate
Political	ICAO	International standards	Indirect	Medium	3	Consider / Consolidate
Economic	Labor unions	Compensation and job security	Indirect	Medium	3	Consider / Consolidate
Technical	Maintenance personnel	Maintainability	Indirect	Medium	3	Consider / Consolidate
Economic	Passengers	Affordable travel	No	High	3	Consider / Consolidate
Economic	Aircraft manufacturers	Product competitiveness	Indirect	Low	2	Boundary input / omit
Environmental	Airport authorities	Sustainable operations	Indirect	Low	2	Boundary input / omit
Technical	Avionics suppliers	System integration	Indirect	Low	2	Boundary input / omit
Social	Cabin crews	Safety and operational effectiveness	Indirect	Low	2	Boundary input / omit
Legal	Certification authorities	Regulatory compliance	No	Medium	2	Boundary input / omit

Table 10 continued

PESTLE domain	Stakeholder	Stated interest	Trajectory impact	Importance	Score	Recommended treatment
Environmental	Climate organizations	Decarbonization	Indirect	Low	2	Boundary input / omit
Social	Families of passengers	Safety	Indirect	Low	2	Boundary input / omit
Environmental	Governments	Emissions targets	Indirect	Low	2	Boundary input / omit
Economic	Insurers	Risk reduction	Indirect	Low	2	Boundary input / omit
Legal	International treaty organizations	Harmonization of standards	Indirect	Low	2	Boundary input / omit
Economic	Investors and shareholders	Return on investment	Indirect	Low	2	Boundary input / omit
Legal	Labor regulators	Worker protections	Indirect	Low	2	Boundary input / omit
Economic	Lessors	Return conditions	Indirect	Low	2	Boundary input / omit
Social	Passengers	Safety and convenience	Indirect	Low	2	Boundary input / omit
Political	State and local governments	Airports, jobs, and noise concerns	No	Medium	2	Boundary input / omit
Social	Tourism industry	Mobility and economic activity	Indirect	Low	2	Boundary input / omit
Technical	Aircraft systems engineers	System performance	No	Low	1	Boundary input / omit
Legal	Aviation lawyers	Compliance and risk	No	Low	1	Boundary input / omit
Technical	Certification engineers	Verification evidence	No	Low	1	Boundary input / omit
Political	Congress and legislatures	Economic growth and public accountability	No	Low	1	Boundary input / omit
Legal	Courts	Liability resolution	No	Low	1	Boundary input / omit
Political	Department of Defense	National security and airspace access	No	Low	1	Boundary input / omit
Political	Department of Transportation	National transportation policy	No	Low	1	Boundary input / omit
Economic	Fuel suppliers	Fuel demand	No	Low	1	Boundary input / omit
Environmental	Future generations (societal proxy)	Long-term sustainability	No	Low	1	Boundary input / omit
Political	National Transportation Safety Board	Prevent accidents	No	Low	1	Boundary input / omit
Legal	Privacy regulators	Passenger data protection	No	Low	1	Boundary input / omit
Technical	Researchers and universities	Innovation	No	Low	1	Boundary input / omit
Technical	Software developers	Correct implementation	No	Low	1	Boundary input / omit
Social	Trainers & Educators	Regulatory stability & standard definitions	No	Low	1	Boundary input / omit

Table 11: Stakeholder objectives traced to requirements: each of the 43 objective rows of the stakeholder objective ontology, the stakeholder requirements that serve it (an airline requirement can serve another stakeholder's objective), and the system requirements and HLRs reached through those that are in the study or held as background.

Objective	**Trajectory impact**	**Stakeholder req. in the study**	**Background or out of study**	**System req. reached (in the study)**	**HLRs reached**
Airline					
Fuel cost	Yes	SN-AIR-01, SN-AIR-05		SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-CLR-13, SLR-INF-02, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-05, HLR-OPS-07, HLR-ATC-05, HLR-ATC-07, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-01, HLR-ACF-05
Crew cost	Indirect	SN-AIR-01		SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-07, HLR-ATC-07, HLR-ACF-01, HLR-ACF-05
Aircraft utilization	Indirect	SN-AIR-02		SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Maintenance	Indirect	SN-AIR-01		SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-07, HLR-ATC-07, HLR-ACF-01, HLR-ACF-05
Schedule integrity	Yes	SN-AIR-02, SN-AIR-03, SN-AIR-05		SLR-CLR-04, SLR-CLR-13, SLR-INF-02, SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-05, HLR-OPS-06, HLR-ATC-05, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-04, HLR-ACF-05, HLR-APT-01, HLR-APT-02
Passenger connections	Yes	SN-AIR-02, SN-AIR-05	SN-AIR-04 (out of study)	SLR-CLR-13, SLR-INF-02, SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-05, HLR-OPS-06, HLR-ATC-05, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-05, HLR-APT-01, HLR-APT-02
Delay cost	Yes	SN-AIR-01, SN-AIR-03	SN-AIR-04 (out of study)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-04, SLR-CLR-09, SLR-INF-07, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-04, HLR-CRW-05, HLR-OPS-01, HLR-OPS-06, HLR-OPS-07, HLR-ATC-07, HLR-ACF-01, HLR-ACF-04, HLR-ACF-05
Revenue	Indirect	SN-AIR-01		SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-07, HLR-ATC-07, HLR-ACF-01, HLR-ACF-05
Dispatch reliability	Indirect	SN-AIR-02		SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Flight Safety	Yes	SN-ATC-01	SN-PAX-01 (out of study)	SLR-PLN-06, SLR-CLR-01, SLR-CLR-02, SLR-INF-01, SLR-EXE-02	HLR-CRW-01, HLR-OPS-02, HLR-ATC-01, HLR-ATC-02, HLR-ATC-03, HLR-ATC-04, HLR-ACF-02, HLR-ACF-06

Table 11 continued

Objective	Trajectory impact	Stakeholder req. in the study	Background or out of study	System req. reached (in the study)	HLRs reached
Passenger					
Ticket price	No	SN-AIR-01		SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-07, HLR-ATC-07, HLR-ACF-01, HLR-ACF-05
Travel time	Yes	SN-AIR-02	SN-PAX-02 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Connection reliability	Yes	SN-AIR-02	SN-PAX-02 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Schedule convenience	Indirect	SN-AIR-01		SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-07, HLR-ATC-07, HLR-ACF-01, HLR-ACF-05
Comfort	Yes	SN-AIR-01	SN-PAX-03 (out of study)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-07, HLR-ATC-07, HLR-ACF-01, HLR-ACF-05
Disruption risk	Indirect	SN-AIR-02	SN-PAX-02 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
ATC / ANSP					
Safety/separation	Yes	SN-ATC-01		SLR-PLN-06, SLR-CLR-01, SLR-CLR-02, SLR-INF-01, SLR-EXE-02	HLR-CRW-01, HLR-OPS-02, HLR-ATC-01, HLR-ATC-02, HLR-ATC-03, HLR-ATC-04, HLR-ACF-02, HLR-ACF-06
Sector workload	Yes	SN-ATC-04		SLR-CLR-15	HLR-ATC-06
Capacity	Yes	SN-ATC-02		SLR-PLN-06	HLR-OPS-02
Predictability	Yes	SN-AIR-02	SN-ATC-03 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Delay (ATC-attributed)	Yes	SN-AIR-02	SN-ATC-05 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Traffic complexity	Yes	SN-ATC-04		SLR-CLR-15	HLR-ATC-06
Airport					
Gate utilization	Indirect	SN-AIR-02	SN-APT-01 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Runway utilization	Yes	SN-AIR-02	SN-APT-02 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Surface congestion	Yes	SN-AIR-02	SN-APT-03 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Turnaround performance	Indirect	SN-AIR-02	SN-APT-01 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02

Table 11 continued

Objective	Trajectory impact	Stakeholder req. in the study	Background or out of study	System req. reached (in the study)	HLRs reached
Passenger throughput	No	SN-AIR-02	SN-APT-01 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Flight Crew					
Safety	Yes	SN-CRW-03, SN-CRW-06, SN-CRW-07	SN-CRW-01 (out of study), SN-CRW-02 (out of study)	SLR-PLN-07, SLR-CLR-01, SLR-CLR-02, SLR-CLR-10, SLR-INF-02, SLR-EXE-03, SLR-EXE-06	HLR-CRW-02, HLR-CRW-03, HLR-CRW-04, HLR-CRW-05, HLR-OPS-03, HLR-OPS-05, HLR-OPS-06, HLR-ATC-01, HLR-ATC-02, HLR-ATC-04, HLR-ATC-08, HLR-ACF-01, HLR-ACF-03, HLR-ACF-05, HLR-ACF-06, HLR-APT-02
Workload	Yes	SN-CRW-03, SN-CRW-08		SLR-CLR-04, SLR-CLR-10, SLR-INF-02, SLR-EXE-03, SLR-EXE-06	HLR-CRW-02, HLR-CRW-03, HLR-CRW-04, HLR-CRW-05, HLR-OPS-05, HLR-OPS-06, HLR-ATC-04, HLR-ATC-08, HLR-ACF-01, HLR-ACF-03, HLR-ACF-04, HLR-ACF-05
Procedural compliance	Yes	SN-CRW-07, SN-CRW-08	SN-CRW-02 (out of study)	SLR-PLN-07, SLR-CLR-04	HLR-CRW-04, HLR-CRW-05, HLR-OPS-03, HLR-OPS-06, HLR-ACF-04, HLR-APT-02
Schedule/duty constraints	Indirect	SN-AIR-02	SN-CRW-05 (out of study)	SLR-INF-07, SLR-APT-04	HLR-CRW-04, HLR-OPS-06, HLR-APT-01, HLR-APT-02
Operational flexibility	Yes	SN-AIR-01, SN-AIR-05	SN-CRW-01 (out of study), SN-CRW-04 (out of study)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-CLR-13, SLR-INF-02, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-05, HLR-OPS-07, HLR-ATC-05, HLR-ATC-07, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-01, HLR-ACF-05
Environmental / Societal					
CO_2	Yes	SN-AIR-01, SN-AIR-05	SN-ENV-01 (background)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-CLR-13, SLR-INF-02, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-05, HLR-OPS-07, HLR-ATC-05, HLR-ATC-07, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-01, HLR-ACF-05
NOx	Yes	SN-AIR-01, SN-AIR-05	SN-ENV-01 (background)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-CLR-13, SLR-INF-02, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-05, HLR-OPS-07, HLR-ATC-05, HLR-ATC-07, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-01, HLR-ACF-05
Contrail/climate effects	Yes	SN-AIR-01, SN-AIR-05	SN-ENV-01 (background)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-CLR-13, SLR-INF-02, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-05, HLR-OPS-07, HLR-ATC-05, HLR-ATC-07, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-01, HLR-ACF-05

Table 11 continued

Objective	Trajectory impact	Stakeholder req. in the study	Background or out of study	System req. reached (in the study)	HLRs reached
Noise	Yes	SN-AIR-01, SN-AIR-05	SN-ENV-02 (out of study)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-CLR-13, SLR-INF-02, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-05, HLR-OPS-07, HLR-ATC-05, HLR-ATC-07, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-01, HLR-ACF-05
Local air quality	Indirect	SN-AIR-01, SN-AIR-05	SN-ENV-01 (background)	SLR-PLN-01, SLR-PLN-02, SLR-CLR-09, SLR-CLR-13, SLR-INF-02, SLR-EXE-03, SLR-MEA-02	HLR-CRW-03, HLR-CRW-05, HLR-OPS-01, HLR-OPS-05, HLR-OPS-07, HLR-ATC-05, HLR-ATC-07, HLR-ATC-08, HLR-ATC-10, HLR-ATC-11, HLR-ACF-01, HLR-ACF-05
Military					
Mission effectiveness	Yes		*none*		
Mission timing	Yes		*none*		
Fuel availability	Indirect		*none*		
Airspace access	Yes		SN-MIL-01 (out of study)		
Operational security	No		*none*		
Resilience	Indirect		*none*		
Stakeholder requirements that name no objective row					
Airline			SN-AIR-06 (out of study)		
Regulator			SN-GOV-01 (background)		
Regulator			SN-GOV-02 (out of study)		

A.3 Requirements Decomposition

Plan ref: to-do-list.md S7, S10 — Introduce the decomposition (Table 12) and cite it from the body: how to read it, why out-of-study requirements stop at the system tier (D-031), and where the studied thread runs (SN-AIR-05 → SLR-INF-02 / SLR-CLR-13).

Table 12: Requirements decomposition: each stakeholder requirement, the system requirements that answer it, and the sub-system requirements (HLRs) that come from each. 30 stakeholder requirements, 42 system requirements (23 in the study) and 31 HLRs, from the SysML v2 model. A requirement's text is given where it first appears; later appearances give its ID. Grey: out of study, with no HLR, the study's cut being made at the HLR tier. [BACKGROUND]: held fixed as an input. [IMPROVED]: not done by today's NAS. [GOAL]: written with "should".

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
Stakeholder: Airline		
SN-AIR-01 The NAS shall provide a means for the airline to plan and fly each flight on the route, altitude and speed it chooses.	**SLR-PLN-01** The NAS shall produce for each scheduled flight a flight plan that states its route, cruise altitude, cruise speed and fuel load. *(also SN-CRW-02)*	**HLR-OPS-01** Flight operations shall plan each flight's route, cruise altitude, cruise speed and fuel load from a model of the performance of the aircraft assigned to it, with the fuel reserve the operating rules require. *(also SLR-PLN-02, SLR-PLN-04)*
	SLR-PLN-02 The NAS shall compute each flight plan from the performance of the aircraft assigned to the flight.	**HLR-OPS-01** *(text above)*
	SLR-CLR-09 The NAS shall answer each flight crew request for a route, altitude or speed change with an approval, an amended clearance or a refusal. *(also SN-CRW-04, SN-PAX-03)*	**HLR-CRW-03** The flight crew shall ask air traffic control for a change of route, altitude or speed when the crew deems it beneficial, optionally in collaboration with the dispatcher. *(also SLR-EXE-03)* **HLR-ATC-07** Air traffic control shall answer each flight crew request for a route, altitude or speed change with an approval, an amended clearance or 'unable'.
	SLR-EXE-03 The NAS shall predict each flight's arrival time, fuel at destination, optimum altitude and maximum altitude from the aircraft's estimated performance. *(also SN-CRW-03)*	**HLR-CRW-03** *(text above)* **HLR-CRW-05** The flight crew shall enter the load sheet weights and the released cost index into the flight management system before departure. *(also SLR-PLN-07)* **HLR-ACF-01** The flight management system shall predict the flight's arrival time, fuel at the destination, optimum altitude and maximum altitude from the entered weights, the cost index and the forecast winds. **HLR-ACF-05** The aircraft shall make its current state available for sharing with the party that decides changes to trajectories. *(also SLR-INF-02)*
	SLR-MEA-02 The NAS shall report the fuel burned and the CO_2 emitted by each flight, at 3.16 kg of CO_2 per kg of fuel. *(also SN-ENV-01)*	**HLR-OPS-07** Flight operations shall report each flight's fuel burned and its CO_2 emitted at 3.16 kg of CO_2 per kg of fuel.

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
SN-AIR-02 The NAS shall provide a means for each flight to achieve arrival at its destination gate at its scheduled arrival time.	**SLR-INF-07** The NAS shall send the airline and the destination airport an updated arrival estimate whenever a flight's estimate changes by more than 5 minutes. *(also SN-AIR-03, SN-APT-01)*	**HLR-CRW-04** The flight crew shall tell the dispatcher of each change to the cleared trajectory. *(also SLR-EXE-06, SLR-CLR-04)* **HLR-OPS-06** The dispatcher shall re-plan the flight with its crew when a trajectory change puts the arrival time, the fuel at the destination or flight safety at risk. *(also SLR-CLR-04, SLR-EXE-06)*
	SLR-APT-01 [OUT OF STUDY] The NAS shall assign each arriving flight a gate before it lands. *(also SN-APT-01)*	*no HLR: not carried into the study*
	SLR-APT-04 The NAS shall have each aircraft fuelled, loaded and sealed by its scheduled off-block time.	**HLR-APT-01** Airport operations shall have each aircraft fuelled, loaded and sealed by its scheduled off-block time. **HLR-APT-02** Airport operations shall report the fuel uplifted and the load put on board to the flight crew and the load planner before off-block. *(also SLR-PLN-07)*
	SLR-MEA-03 [OUT OF STUDY] The NAS shall report each flight's gate arrival time against its scheduled arrival time. *(also SN-PAX-02)*	*no HLR: not carried into the study*
SN-AIR-03 The NAS shall inform the airline of airspace and airport constraints that will delay or reroute its flights early enough for the airline to re-plan.	**SLR-FLW-07** [OUT OF STUDY] The NAS shall give traffic flow managers and airlines the activation schedule of special use airspace before the affected flight plans are filed. *(also SN-MIL-01)*	*no HLR: not carried into the study*
	SLR-CLR-04 The NAS shall provide a two-way communication path from the flight crew to the controlling facility and to the dispatcher for the whole flight. *(also SN-CRW-08)*	**HLR-CRW-04** *(text above)* **HLR-OPS-06** *(text above)* **HLR-ACF-04** The aircraft communication system shall carry voice and data between the flight crew and both air traffic control and the dispatcher for the whole flight.
	SLR-INF-07 *(text above)*	**HLR-CRW-04** *(text above)* **HLR-OPS-06** *(text above)*

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
SN-AIR-04 [OUT OF STUDY] The NAS shall let the airline choose which of its own flights takes a delay that the airline has been assigned.	**SLR-FLW-04** [OUT OF STUDY] The NAS shall let an airline exchange its own flights among the departure slots it has been assigned.	*no HLR: not carried into the study*
SN-AIR-05 The NAS shall take the airline's priorities for a flight (fuel against time, and protected connections) into account when a change to that flight's trajectory is required.	**SLR-CLR-13** [IMPROVED] The NAS shall choose, among the maneuvers that restore separation, the one with the lowest impact on the aircraft involved. *(also SN-ENV-01)*	**HLR-ATC-05** [IMPROVED] Air traffic control shall choose, among the viable resolution options, the one with the lowest impact on the systems involved. **HLR-ATC-10** [IMPROVED] ATC decision support shall determine, for each predicted conflict, every resolution option that restores separation and that each aircraft it moves can fly. **HLR-ATC-11** [IMPROVED] ATC decision support shall determine the impact of each viable resolution option on the systems involved.
	SLR-INF-02 [IMPROVED] The NAS shall give the party that decides a change to an aircraft's trajectory the information needed to preserve safety and pursue optimality. *(also SN-CRW-03)*	**HLR-OPS-05** [IMPROVED] Flight operations shall make each flight's current aircraft state and performance, and the airline's priorities for it, available to the party that decides a change to that flight's trajectory. **HLR-ATC-08** [IMPROVED] ATC decision support shall use the best information available to it about each flight, including what the flight's airline shares. **HLR-ACF-05** *(text above)*
SN-AIR-06 [OUT OF STUDY] The NAS shall keep the airline's commercially sensitive flight data (aircraft weight, cost index, connection priorities) private.	**SLR-INF-05** [OUT OF STUDY] The NAS shall disclose an airline's shared weight, cost index and connection priority only to the parties deciding that flight's trajectory.	*no HLR: not carried into the study*
Stakeholder: Air traffic control		
SN-ATC-01 The NAS shall keep every controlled aircraft separated from every other aircraft by at least the applicable separation minimum.	**SLR-PLN-06** The NAS shall file each flight plan with air traffic control before the flight departs. *(also SN-ATC-02, SN-ATC-03)*	**HLR-OPS-02** The dispatcher shall file each flight's plan with air traffic control before departure.

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
	SLR-CLR-01 The NAS shall keep each pair of controlled aircraft separated by at least the minimum that JO 7110.65BB sets for the airspace they are in. *(also SN-CRW-06, SN-PAX-01)*	**HLR-ATC-01** Air traffic control shall keep each pair of aircraft it controls separated by at least the applicable minimum, allowing for each aircraft's permitted deviation from its cleared route, altitude and speed. *(also SLR-EXE-02)* **HLR-ATC-04** Air traffic control shall resolve each predicted conflict with a clearance that every aircraft it moves can fly. *(also SLR-CLR-10)* **HLR-ACF-06** The aircraft's surveillance system shall broadcast the aircraft's position, altitude, velocity and identity by ADS-B Out. *(also SLR-CLR-02)*
	SLR-CLR-02 The NAS shall predict a loss of separation between controlled aircraft before it occurs. *(also SN-CRW-06)*	**HLR-ATC-02** ATC decision support shall alert the sector controller to each predicted loss of separation before it occurs. **HLR-ACF-06** *(text above)*
	SLR-CLR-03 [OUT OF STUDY] The NAS shall give the controlling facility the position, altitude and identity of each controlled aircraft. *(also SN-ATC-03)*	*no HLR: not carried into the study*
	SLR-CLR-05 [OUT OF STUDY] The NAS shall change a controlled aircraft's cleared route, altitude or speed only by a clearance from the facility that has control of that aircraft. *(also SN-GOV-02)*	*no HLR: not carried into the study*
	SLR-INF-01 The NAS shall give the sector controller the filed route, requested altitude and requested speed of each aircraft before it enters the sector. *(also SN-ATC-03)*	**HLR-ATC-03** The en-route center shall inform the sector controller of each aircraft's filed route, requested altitude and requested speed before the aircraft enters the sector.
	SLR-EXE-02 The NAS shall fly each aircraft along its cleared route and at its cleared altitude and speed within the navigation and altitude tolerances of the airspace. *(also SN-ATC-03)*	**HLR-CRW-01** The flight crew shall safely control the flight from origin to destination, as cleared by air traffic control with or without negotiation, and in collaboration with flight operations. *(also SLR-CLR-08, SLR-PLN-05)* **HLR-ATC-01** *(text above)* **HLR-ACF-02** The auto flight control system shall hold the aircraft on its cleared route, altitude and speed within the tolerances of the airspace.

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
SN-ATC-02 The NAS shall keep the number of aircraft in each sector and at each airport within its declared capacity.	**SLR-PLN-06** *(text above)*	**HLR-OPS-02** *(text above)*
	SLR-PLN-08 [OUT OF STUDY] The NAS shall plan each flight to comply with the traffic management initiatives and special use airspace activations published for its time of flight. *(also SN-MIL-01)*	*no HLR: not carried into the study*
	SLR-FLW-06 [OUT OF STUDY] The NAS shall release each flight that has an expect departure clearance time (EDCT) within the permitted window around that time. *(also SN-ATC-03)*	*no HLR: not carried into the study*
	SLR-APT-02 [OUT OF STUDY] The NAS shall sequence arrivals and departures so that each runway is used at its declared capacity whenever demand is at or above that rate. *(also SN-APT-02)*	*no HLR: not carried into the study*
SN-ATC-03 [OUT OF STUDY] The NAS shall make the path each controlled aircraft will fly known to its controller in advance.	**SLR-PLN-06** *(text above)*	**HLR-OPS-02** *(text above)*
	SLR-FLW-06 *(text above)*	*no HLR: not carried into the study*
	SLR-CLR-03 *(text above)*	*no HLR: not carried into the study*
	SLR-INF-01 *(text above)*	**HLR-ATC-03** *(text above)*
	SLR-EXE-01 [OUT OF STUDY] The NAS shall hold each flight's cleared route and altitudes in the flight management system as the path the aircraft's guidance follows.	*no HLR: not carried into the study*
	SLR-EXE-02 *(text above)*	**HLR-CRW-01** *(text above)* **HLR-ATC-01** *(text above)* **HLR-ACF-02** *(text above)*
SN-ATC-04 The NAS shall keep the number of control actions a sector controller must take within what one sector team can sustain.	**SLR-CLR-15** The NAS shall keep the control instructions needed to resolve a conflict within what the sector team can sustain.	**HLR-ATC-06** Air traffic control shall resolve each predicted conflict within the number of instructions the sector team can sustain.

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
SN-ATC-05 [OUT OF STUDY] The NAS shall share unavoidable delay equitably among airspace users.	**SLR-FLW-05** [OUT OF STUDY] The NAS shall assign the delay created by a traffic management initiative equitably among all airspace users.	*no HLR: not carried into the study*
	SLR-MEA-04 [OUT OF STUDY] The NAS shall report the delay each flight received from traffic management initiatives, by airline.	*no HLR: not carried into the study*
Stakeholder: Flight crew		
SN-CRW-01 [OUT OF STUDY] The NAS shall preserve the captain's final authority over the safe conduct of the flight.	**SLR-PLN-05** The NAS shall release a flight only when the dispatcher and the captain both agree that it can be flown safely as planned. *(also SN-GOV-01, SN-GOV-02)*	**HLR-CRW-01** *(text above)*
		HLR-OPS-04 The dispatcher shall release a flight only with the captain's agreement that it can be flown safely as planned.
	SLR-CLR-08 The NAS shall allow the captain to deviate from a clearance when safety requires it, with air traffic control told as soon as practical. *(also SN-GOV-01, SN-PAX-01)*	**HLR-CRW-01** *(text above)*
	SLR-EXE-04 [OUT OF STUDY] The NAS shall change an aircraft's flown trajectory only through flight deck controls operated by the flight crew. *(also SN-GOV-02)*	*no HLR: not carried into the study*
SN-CRW-02 [OUT OF STUDY] The NAS shall give the flight crew the information needed for each phase of flight before that phase begins.	**SLR-PLN-01** *(text above)*	**HLR-OPS-01** *(text above)*
	SLR-PLN-07 The NAS shall deliver the dispatch release, flight plan, weather, notices to airmen (NOTAMs) and load sheet to the flight crew before departure. *(also SN-CRW-07)*	**HLR-CRW-05** *(text above)*
		HLR-OPS-03 Flight operations shall give the flight crew the dispatch release, flight plan, weather, notices to airmen (NOTAMs) and load sheet before departure. **HLR-APT-02** *(text above)*

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
	SLR-INF-06 [OUT OF STUDY] The NAS shall give the dispatcher and the flight crew the reported and forecast weather and turbulence along the planned route. *(also SN-PAX-03)*	*no HLR: not carried into the study*
	SLR-EXE-06 The NAS shall alert the flight crew, dispatchers and controllers when the fuel predicted at the destination falls below the required reserve. *(also SN-GOV-01, SN-CRW-03, SN-PAX-01)*	**HLR-CRW-04** *(text above)* **HLR-OPS-06** *(text above)* **HLR-ACF-03** The flight management system shall alert the flight crew when the fuel it predicts at the destination falls below the required reserve.
SN-CRW-03 The NAS shall give the flight crew only trajectory clearances and instructions that the aircraft can fly, with enough time to set them up.	**SLR-CLR-10** The NAS shall issue only clearances that the aircraft can fly.	**HLR-CRW-02** The flight crew shall accept a clearance only if the aircraft can fly it, and shall answer 'unable' when it cannot. **HLR-ATC-04** *(text above)*
	SLR-INF-02 *(text above)*	**HLR-OPS-05** *(text above)* **HLR-ATC-08** *(text above)* **HLR-ACF-05** *(text above)*
	SLR-EXE-03 *(text above)*	**HLR-CRW-03** *(text above)* **HLR-CRW-05** *(text above)* **HLR-ACF-01** *(text above)* **HLR-ACF-05** *(text above)*
	SLR-EXE-06 *(text above)*	**HLR-CRW-04** *(text above)* **HLR-OPS-06** *(text above)* **HLR-ACF-03** *(text above)*
SN-CRW-04 [OUT OF STUDY] The NAS shall let the flight crew ask for a trajectory change and get an answer.	**SLR-CLR-09** *(text above)*	**HLR-CRW-03** *(text above)* **HLR-ATC-07** *(text above)*
SN-CRW-05 [OUT OF STUDY] The NAS shall keep each flight crew member within flight and duty time limits.	**SLR-PLN-09** [OUT OF STUDY] The NAS shall assign to each flight a crew whose duty period, including forecast delay, stays within the limits of 14 CFR Part 117. *(also SN-GOV-01)*	*no HLR: not carried into the study*

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
SN-CRW-06 The NAS shall provide the flight crew clearances that account for air traffic.	**SLR-CLR-01** *(text above)*	**HLR-ATC-01** *(text above)*
		HLR-ATC-04 *(text above)* **HLR-ACF-06** *(text above)*
	SLR-CLR-02 *(text above)*	**HLR-ATC-02** *(text above)* **HLR-ACF-06** *(text above)*
SN-CRW-07 The NAS shall inform the flight crew of weather, traffic and other safety considerations.	**SLR-PLN-07** *(text above)*	**HLR-CRW-05** *(text above)*
		HLR-OPS-03 *(text above)* **HLR-APT-02** *(text above)*
SN-CRW-08 The NAS shall facilitate control of the aircraft across regional sectors, coordinating controls, clearances and handoffs.	**SLR-CLR-04** *(text above)*	**HLR-CRW-04** *(text above)*
		HLR-OPS-06 *(text above)* **HLR-ACF-04** *(text above)*
Stakeholder: Regulator		
SN-GOV-01 [BACKGROUND] The NAS shall conduct every flight within the operating rules of 14 CFR and the FAA's air traffic orders.	**SLR-PLN-04** The NAS shall load each flight with fuel to reach its destination, then its most distant alternate, then fly 45 minutes at normal cruise. *(also SN-PAX-01)*	**HLR-OPS-01** *(text above)*
	SLR-PLN-05 *(text above)*	**HLR-CRW-01** *(text above)* **HLR-OPS-04** *(text above)*
	SLR-PLN-09 *(text above)*	*no HLR: not carried into the study*
	SLR-CLR-08 *(text above)*	**HLR-CRW-01** *(text above)*
	SLR-EXE-06 *(text above)*	**HLR-CRW-04** *(text above)* **HLR-OPS-06** *(text above)* **HLR-ACF-03** *(text above)*
	SLR-MEA-01 The NAS shall record each change to an aircraft's trajectory, to its clearance, or to the constraints imposed on it, with the authority that issued it and the time. *(also SN-GOV-02)*	**HLR-ATC-09** Air traffic control shall record each clearance it issues with the issuing facility and the time.

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
SN-GOV-02 [OUT OF STUDY] The NAS shall make one identified authority accountable for each decision that changes a flight's trajectory.	**SLR-PLN-05** *(text above)*	**HLR-CRW-01** *(text above)*
		HLR-OPS-04 *(text above)*
	SLR-CLR-05 *(text above)*	*no HLR: not carried into the study*
	SLR-EXE-04 *(text above)*	*no HLR: not carried into the study*
	SLR-MEA-01 *(text above)*	**HLR-ATC-09** *(text above)*
Stakeholder: Communities and environment		
SN-ENV-01 [BACKGROUND] The NAS shall keep the CO_2 emitted by each flight as low as safety, economics and capacity allow.	**SLR-CLR-13** *(text above)*	**HLR-ATC-05** *(text above)*
		HLR-ATC-10 *(text above)* **HLR-ATC-11** *(text above)*
	SLR-APT-03 [OUT OF STUDY] The NAS shall hold a flight that has a known departure delay at its gate with engines off, not in a taxiway queue. *(also SN-APT-03)*	*no HLR: not carried into the study*
	SLR-MEA-02 *(text above)*	**HLR-OPS-07** *(text above)*
SN-ENV-02 [OUT OF STUDY] The NAS shall keep aircraft noise over communities near airports within the published noise abatement procedures.	**SLR-APT-05** [OUT OF STUDY] The NAS shall assign departure and arrival procedures that follow the airport's published noise abatement procedures.	*no HLR: not carried into the study*
Stakeholder: Airport operator		
SN-APT-01 [OUT OF STUDY] The NAS shall give the airport operator arrival and departure times reliable enough to assign gates and ground crews.	**SLR-INF-07** *(text above)*	**HLR-CRW-04** *(text above)*
		HLR-OPS-06 *(text above)*
	SLR-APT-01 *(text above)*	*no HLR: not carried into the study*
SN-APT-02 [OUT OF STUDY] The NAS shall use the runway capacity the airport has declared available.	**SLR-APT-02** *(text above)*	*no HLR: not carried into the study*

Table 12 continued

Stakeholder requirement	System requirement (the NAS shall)	Sub-system requirement (HLR)
SN-APT-03 [OUT OF STUDY] The NAS shall keep the time aircraft wait on taxiways with engines running as short as practical.	**SLR-APT-03** *(text above)*	*no HLR: not carried into the study*
Stakeholder: Passengers		
SN-PAX-01 [OUT OF STUDY] The NAS shall carry passengers without injury.	**SLR-PLN-04** *(text above)*	**HLR-OPS-01** *(text above)*
	SLR-CLR-01 *(text above)*	**HLR-ATC-01** *(text above)* **HLR-ATC-04** *(text above)* **HLR-ACF-06** *(text above)*
	SLR-CLR-08 *(text above)*	**HLR-CRW-01** *(text above)*
	SLR-EXE-06 *(text above)*	**HLR-CRW-04** *(text above)* **HLR-OPS-06** *(text above)* **HLR-ACF-03** *(text above)*
SN-PAX-02 [OUT OF STUDY] The NAS shall get passengers to their destination, including connections, at the time they were sold.	**SLR-MEA-03** *(text above)*	*no HLR: not carried into the study*
SN-PAX-03 [OUT OF STUDY] The NAS shall avoid known turbulence where a route or altitude change allows it.	**SLR-CLR-09** *(text above)*	**HLR-CRW-03** *(text above)* **HLR-ATC-07** *(text above)*
	SLR-INF-06 *(text above)*	*no HLR: not carried into the study*
Stakeholder: Military airspace users		
SN-MIL-01 [OUT OF STUDY] The NAS shall give military operations sole use of special use airspace while it is scheduled active.	**SLR-PLN-08** *(text above)*	*no HLR: not carried into the study*
	SLR-FLW-07 *(text above)*	*no HLR: not carried into the study*
	SLR-CLR-14 [OUT OF STUDY] The NAS shall keep civil aircraft out of special use airspace while it is active, unless the using agency releases it.	*no HLR: not carried into the study*

A.4 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in the body: full BDD/IBD set, requirements model, and complete traceability matrices (needs → objectives → requirements → systems → activities → information exchanges → aircraft behavior). Not yet modeled as views: BDDs and the requirements model.

Every figure of the model in this report is a view defined in `nas_package_my_views.sysml` and rendered from the SysML v2 text with Syside. Its caption names the view, the `.sysml` file that holds the content, and the date of the render. As of this draft all five views appear in the body: Figure 4 (Section 3.3), Figure 5 (Section 3.4), and Figures 8, 9 and 19 (Section 3.5). Views added to the model later appear below.

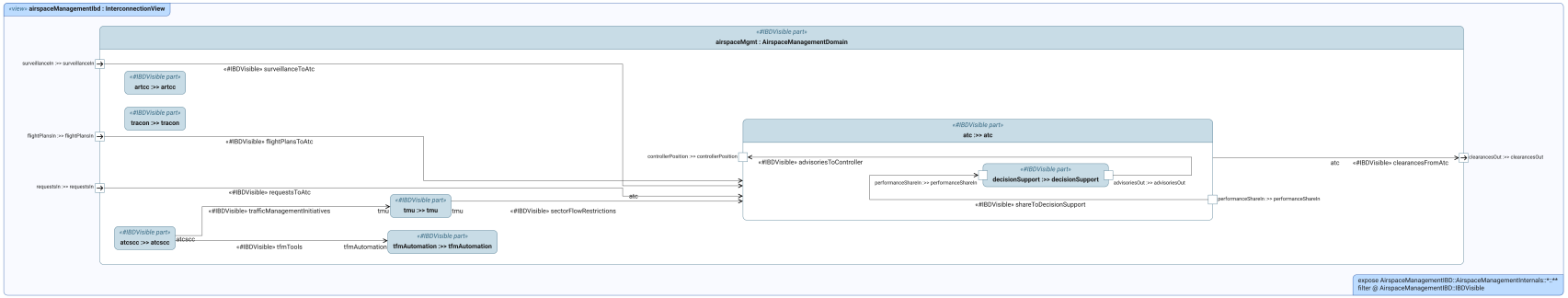


Figure 19: Airspace management internal block diagram. View `airspaceManagementIbd`; source `nas_ibd_airspace_management.sysml`; rendered 2026-11-05.

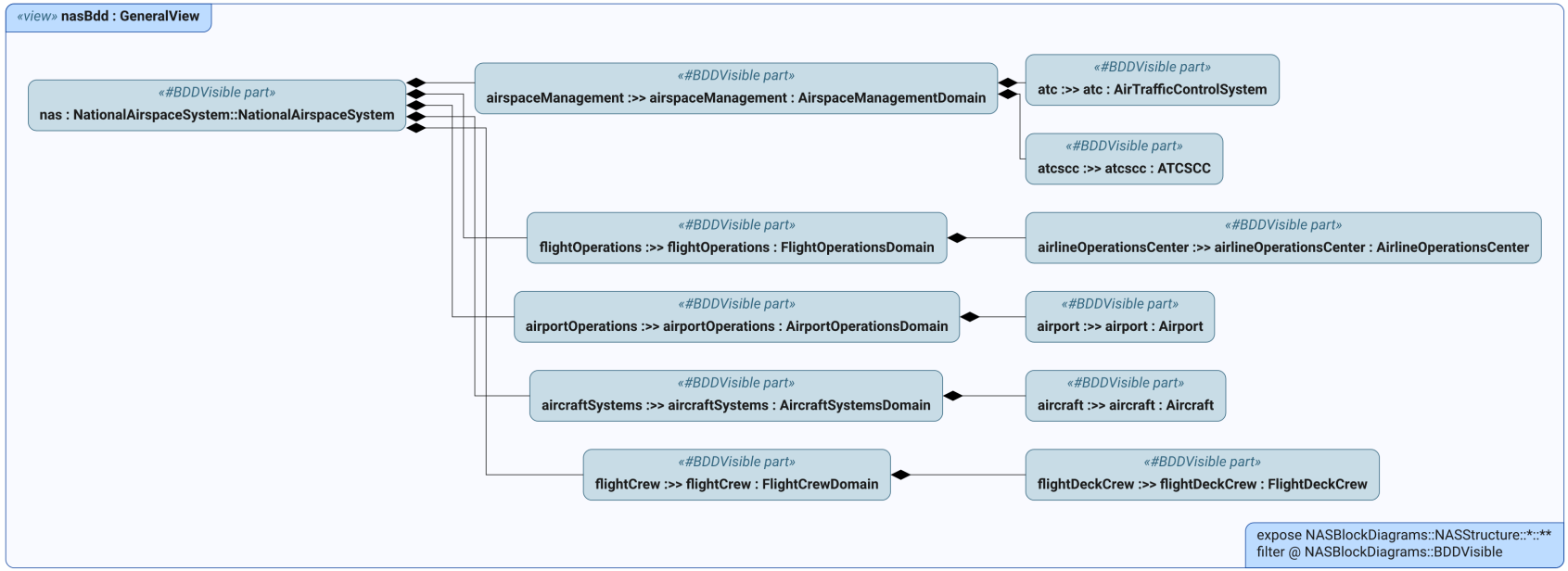


Figure 20: NAS block definition diagram: the modeled domains and the context-diagram systems, composition only. View `nasBdd`; source `nas_block_diagrams.sysml`; rendered 2026-10-19.

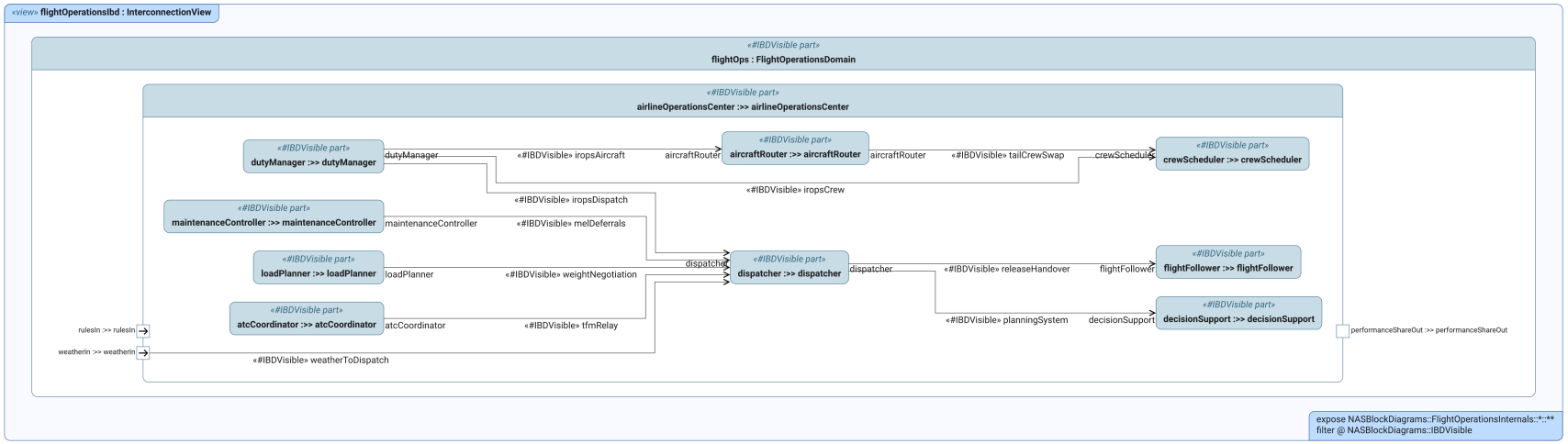


Figure 21: Flight operations internal block diagram: the airline operations center's roles. View `flightOperationsIbd`; source `nas_block_diagrams.sysml`; rendered 2026-10-19.

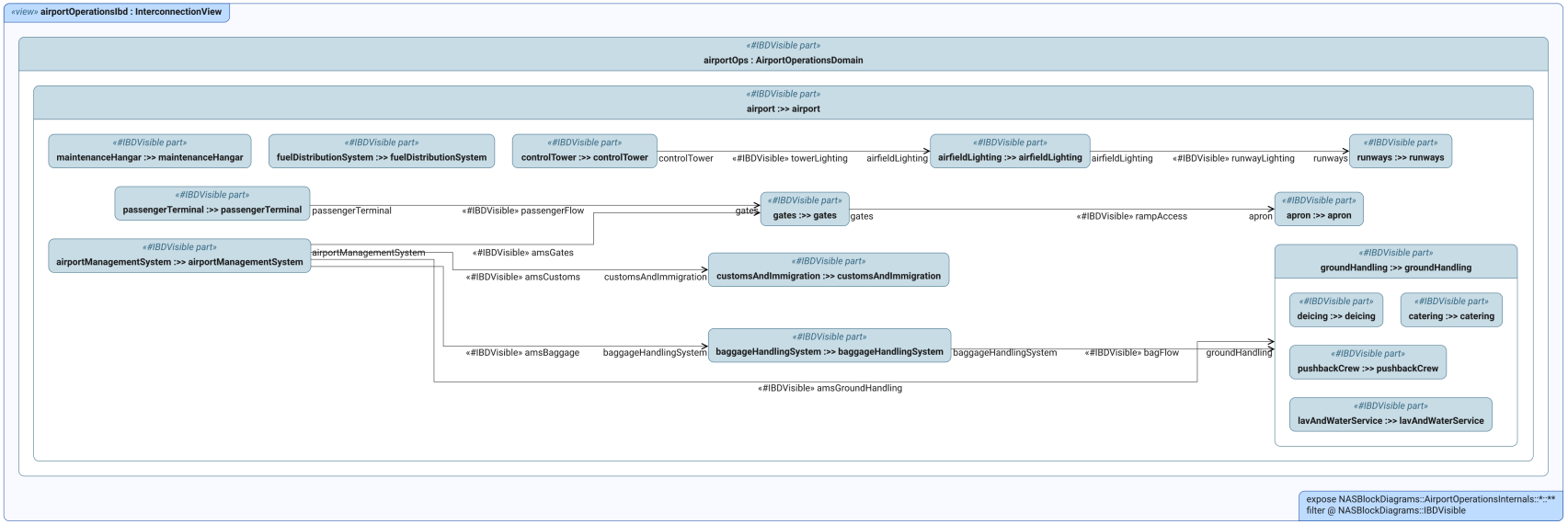


Figure 22: Airport operations internal block diagram. View `airportOperationsIbd`; source `nas_block_diagrams.sysml`; rendered 2026-10-19.

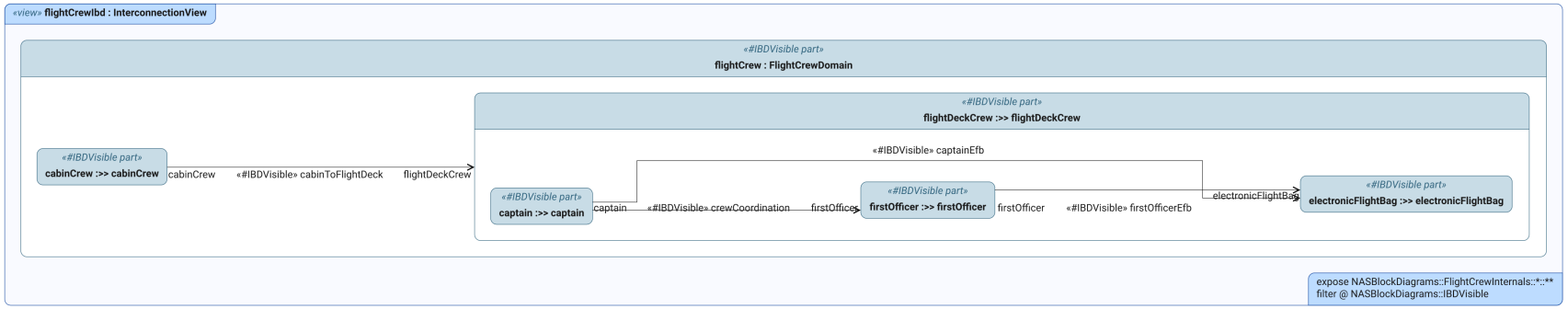


Figure 23: Flight crew internal block diagram. View `flightCrewIbd`; source `nas_block_diagrams.sysml`; rendered 2026-10-19.

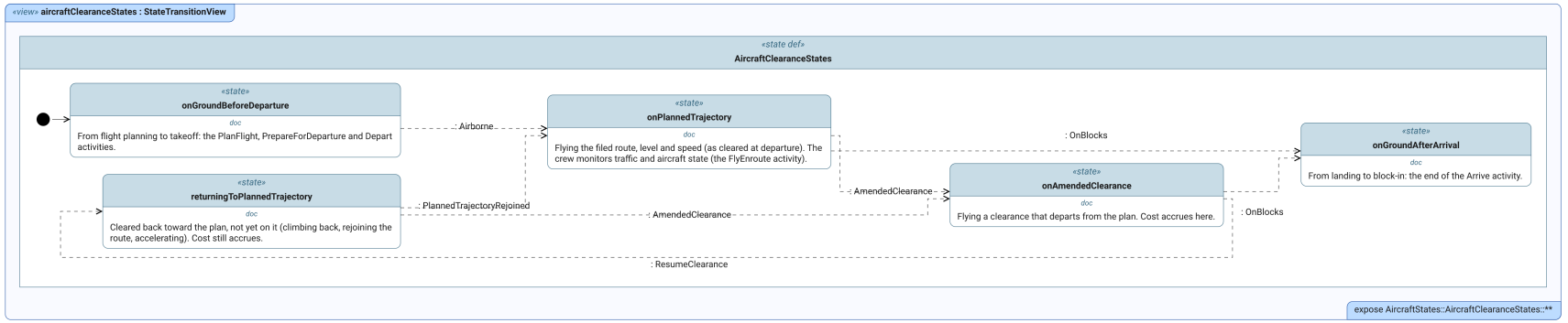


Figure 24: Aircraft state machine: the clearance the aircraft is flying, from the ground before departure to the ground after arrival. View `aircraftClearanceStates`; source `nas_state_aircraft.sysml`; rendered 2026-11-05.

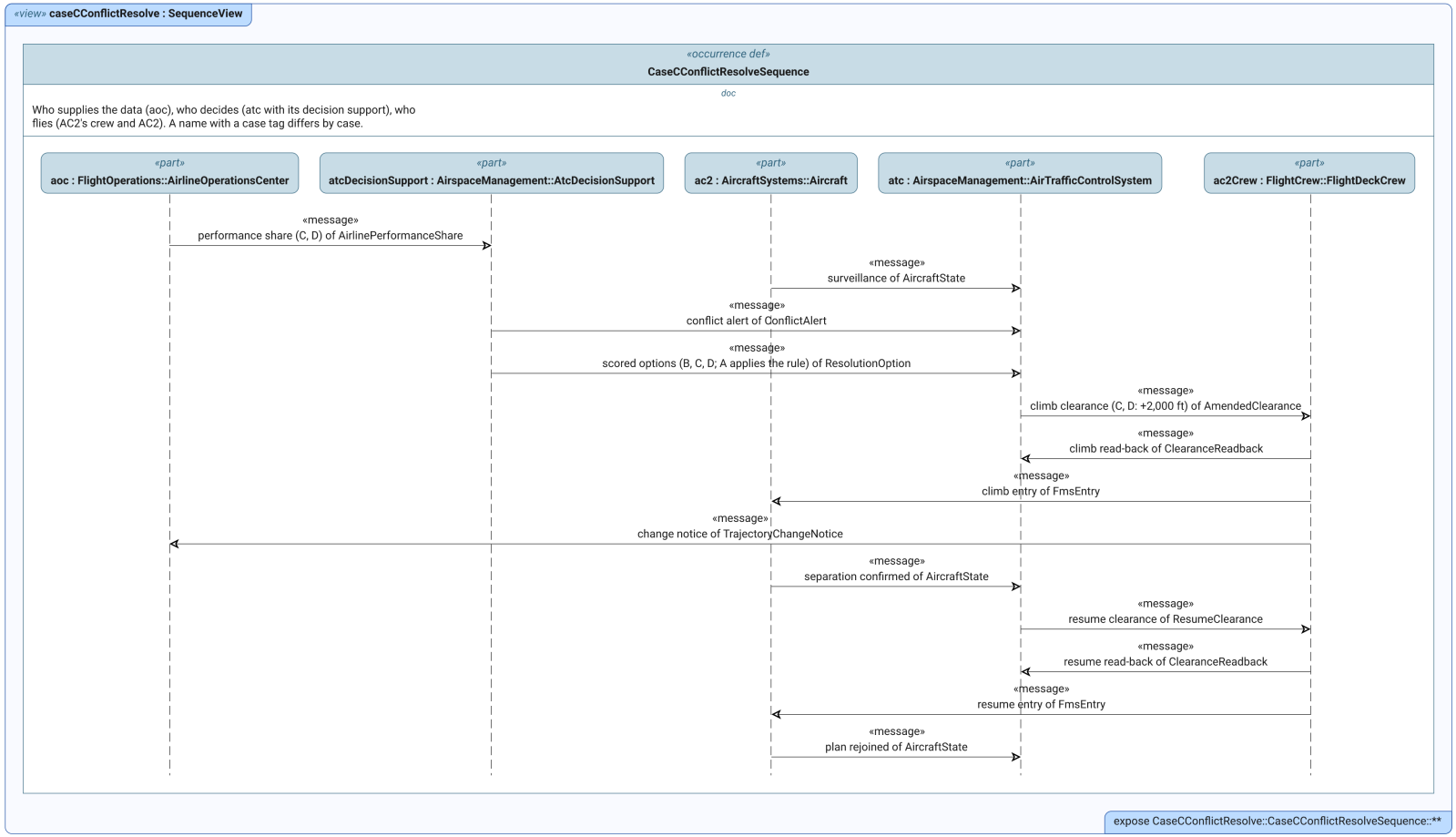


Figure 25: Case C Conflict Resolve: the exchanges when ATC decision support scores the options with the airline's shared data. View `caseCConflictResolve`; source `nas_sequence_case_c.sysml`; rendered 2026-11-05.

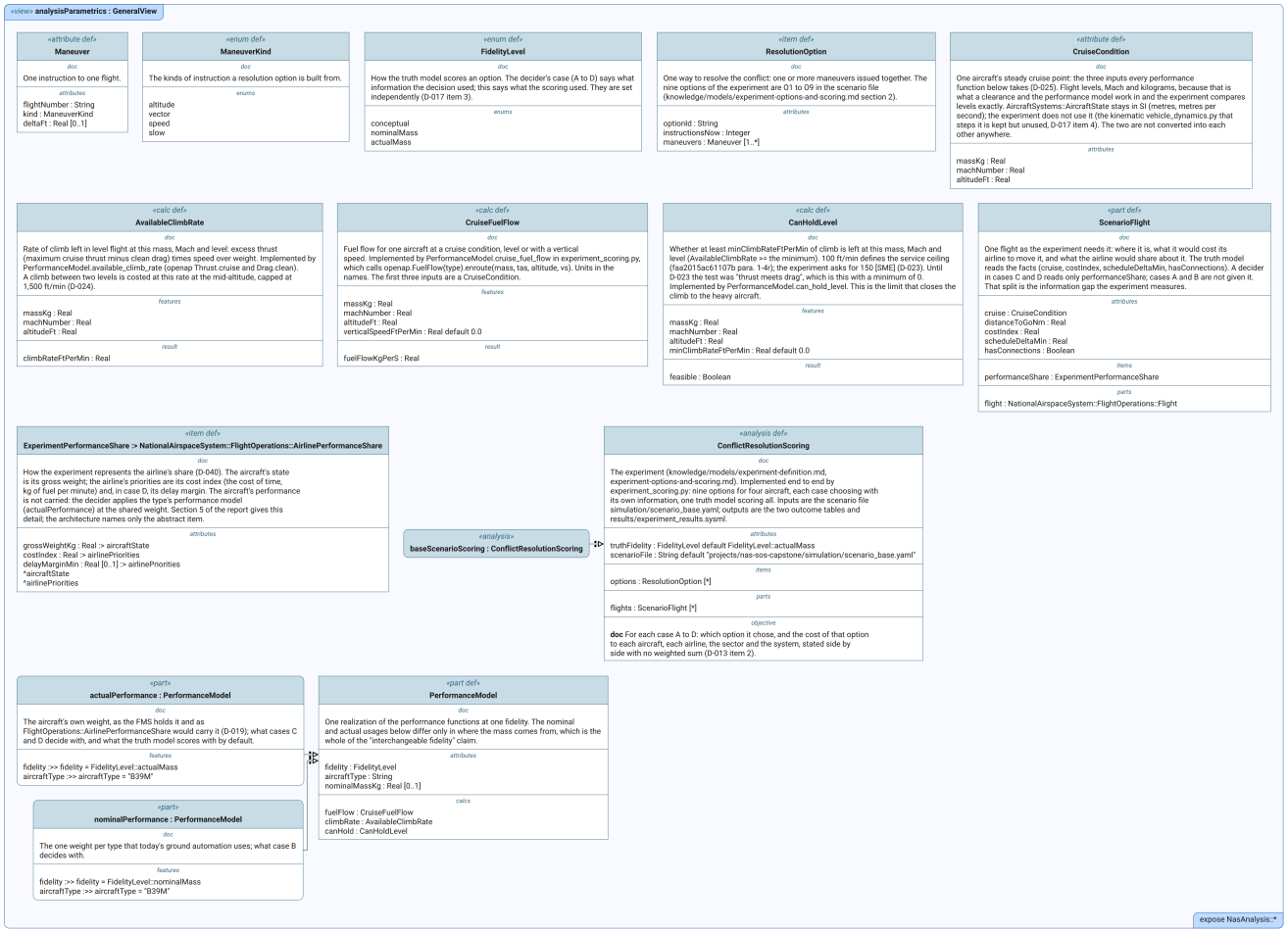


Figure 26: Analysis definitions: the performance calculations, the two fidelity usages and the conflict-resolution scoring analysis. View `analysisParametrics`; source `nas_analysis.sysml`; rendered 2026-11-05.

A.5 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.6 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 5.

Plan ref: to-do-list.md S11 — Full options table: for each of the ten options, the maneuver per aircraft, instructions now, follow-up later, and what the decider must know to use it. Draft in `knowledge/models/experiment-options-and-scoring.md` §2.

Plan ref: to-do-list.md S11, S12 — Every option scored at four levels (the table from which each case picks one row), the per-viewpoint cost statements with their symbols defined, and the table of abstractions with reason and risk.

A.7 Generative AI Use Disclosure

Generative AI was used as an assistant throughout this project. VS Code was the primary environment in which content was created, and the Claude Code extension (Anthropic) [cite] provided guidance, feedback, and draft content within it. Before that, two ChatGPT (OpenAI) [cite] conversations helped shape the project's direction. They were part of the reasoning that moved it from a trajectory-optimization study to the NAS modeled as a system of systems, with optimization as one capability inside the architecture. All AI-drafted content was reviewed and modified as needed to reflect my own views, understanding, and work.

Beyond drafting content, AI was used as a guide in three ways. It supported schedule and progress management against the course milestones. It gave technical guidance. And it critiqued the work, sometimes in deliberately demanding reviewer roles. AI gave initial reviews of all sources I identified and downloaded, to help me prioritize my own reading. After I read and marked up a source, AI extracted my markings and folded them into a library of sources and excerpts that I drew on while writing. As I wrote report sections, I asked AI for reviews, modifications, critiques, and relevant sources and excerpts.

The project's engineering content was produced the same way: AI drafted, and I reviewed, usually by hand, on paper. AI drafted much of the SysML v2 model text and the requirements, using my own work (leveraged from prior classes and authored as part of this project) as a starting point. I printed the diagrams and requirements, marked them up by hand, and AI transcribed my markings and applied them for me to check. The requirements show the effect. AI's first draft had 27 stakeholder and 53 system requirements. Over three hand reviews I struck, reworded, or wrote requirements until the set covered only what bears on the decision studied. Links in the model carry an evidence grade, so a reader can tell a link drawn from a published source or FAA order from one drawn from my industry experience or from general knowledge.

I formulated the experiment, both conceptually and in detail. AI drafted the code that implements it, runs it, and measures its outputs. AI also wrote the supporting scripts: rendering the model's diagrams, exporting the requirements, generating this report's requirement tables, and extracting annotations from PDFs. Every scenario value used in the report was set or confirmed by me.

This disclosure is itself an example of the process. I wrote the first version. AI checked it against the project's decision log and journals and found what it left out: the ChatGPT conversations, the

extent of the model and requirements drafting, and the transcription of my hand reviews. It then drafted this revision, which I edited. As with the rest of the work, the check against the record was the useful part, and the final wording is mine.

A.8 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (`evidence/source-register.md`): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.